

DEPARTMENT OF MATHEMATICS

MODULE 2: FOURIER SERIES

Introduction:

Fourier series are infinite series designed to represent general periodic functions in terms of simple ones, namely sines and cosines. This trigonometric system is orthogonal allowing the computation of the co-efficients of the Fourier series by the use of well known Euler formulas.

Fourier series are very important to engineers and physicists because they allow the solution of ODEs in connection with forced oscillations and the approximations of periodic functions. Fourier series in certain sense are more universal than the more familiar Taylor's series in calculus because many discontinuous periodic functions that come up in applications can be developed in Fourier series but do not have Taylor's series expansions.

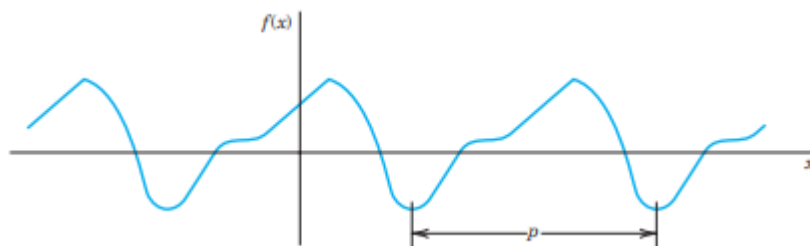
Fourier series can be extended to non-periodic phenomena to obtain Fourier integrals and Fourier transforms, where both play an important application in solving **PDEs**.

In the digital age, the **discrete Fourier transform** plays an important role. Signals, such as voice or music are sampled and analysed. **Fast Fourier transform** is the important algorithm in this context.

Fourier series:

Periodic function: Any function which repeats itself at regular intervals of time is called as a periodic function.

Mathematically, a function $f(x)$, $\forall x$ which is real except possibly at some points is said to be periodic with period p if $f(x + p) = f(x)$, p is some positive number.



The smallest positive period is called *fundamental period*.

Examples of periodic functions:

- 1) Sine
 - 2) Cosine
- (both are periodic functions of period $p = 2\pi$)



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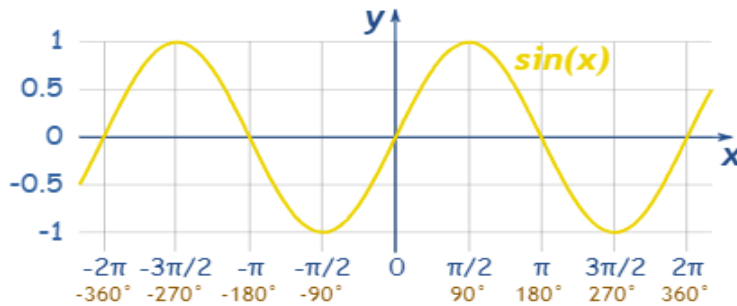


Fig 1: graph of $y = \sin x$

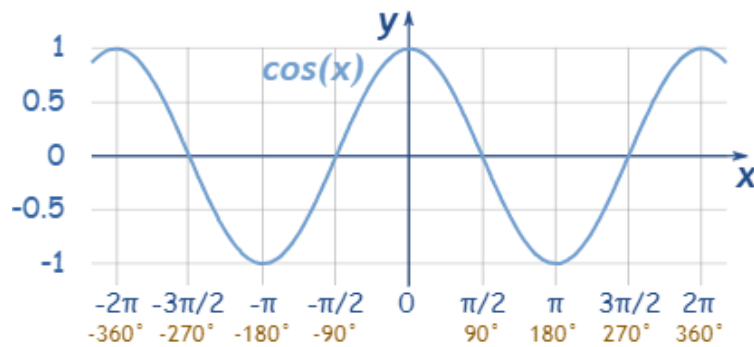
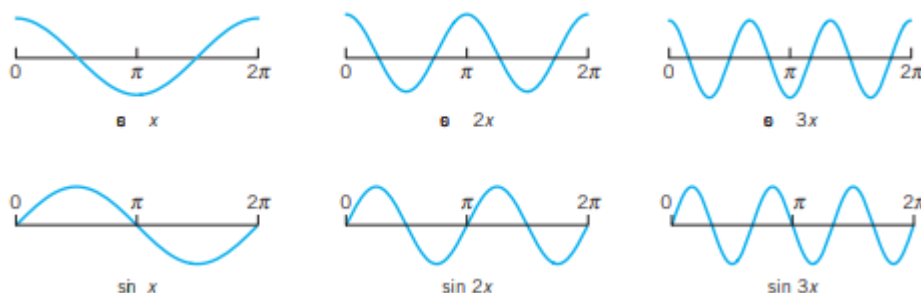


Fig 2: graph of $y = \cos x$

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Representation of various functions of period $p = 2\pi$

$\cos(x)$ & $\sin(x)$, $\cos(2x)$ & $\sin(2x)$, $\cos(3x)$ & $\sin(3x)$,



Note:

- 1) $f(x) = \cos x$ is periodic of period $p = 2\pi$
 $f(x + p) = \cos\{x + p\} = \cos x \cos p - \sin x \sin p$
Let $p = 2\pi \therefore f(x + 2\pi) = \cos x \cos(2\pi) - \sin x \sin(2\pi)$
 $\therefore f(x + 2\pi) = \cos x = f(x)$
- 2) $f(x) = \cos 2x$ is periodic of period $p = \pi$
 $f(x + p) = \cos 2\{x + p\} = \cos\{2x + 2p\} = \cos 2x \cos 2p - \sin 2x \sin 2p$
Let $p = \pi \therefore f(x + \pi) = \cos 2x \cos(2\pi) - \sin 2x \sin(2\pi)$
 $\therefore f(x + \pi) = \cos 2x = f(x) \therefore p = \pi$ is the fundamental period
- 3) $f(x) = \sin 2x$ is periodic of period $p = \pi$
 $f(x + p) = \sin 2\{x + p\} = \sin\{2x + 2p\} = \sin 2x \cos 2p + \cos 2x \sin 2p$
Let $p = \pi \therefore f(x + \pi) = \sin 2x \cos 2\pi + \cos 2x \sin 2\pi$
 $\therefore f(x + \pi) = \sin 2x = f(x) \therefore p = \pi$ is the fundamental period
- 4) $f(x) = \cos 3x$ is periodic of period $p = \frac{2\pi}{3}$
 $f(x + p) = \cos 3\{x + p\} = \cos\{3x + 3p\} = \cos 3x \cos 3p - \sin 3x \sin 3p$
Let $p = \frac{2\pi}{3} \therefore f\left(x + \frac{2\pi}{3}\right) = \cos 3x \cos(2\pi) - \sin 3x \sin(2\pi)$
 $\therefore f(x + \frac{2\pi}{3}) = \cos 3x = f(x) \therefore p = \frac{2\pi}{3}$ is the fundamental period
- 5) $f(x) = \sin 3x$ is periodic of period $p = \frac{2\pi}{3}$
 $f(x + p) = \sin 3\{x + p\} = \sin\{3x + 3p\} = \sin 3x \cos 3p + \cos 3x \sin 3p$

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$$\text{Let } p = \frac{2\pi}{3} \therefore f\left(x + \frac{2\pi}{3}\right) = \sin 3x \cos(2\pi) + \cos 3x \sin(2\pi)$$

$$\therefore f(x + 2\pi) = \sin 3x = f(x) \therefore p = \frac{2\pi}{3} \text{ is the fundamental period}$$

In general any trigonometric function say $\cos(nx)$ & $\sin(nx)$ is periodic of period

$$p = \frac{2\pi}{n}, n = 1, 2, 3, \dots$$

The series to be obtained will be a *trigonometric* series which will be of the form

$$\begin{aligned} & \frac{a_0}{2} + \{a_1 \cos x + b_1 \sin x\} + \{a_2 \cos x + b_2 \sin x\} + \{a_3 \cos x + b_3 \sin x\} + \dots \\ & = \frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_n \cos(nx) + b_n \sin(nx)\} \end{aligned} \quad \dots\dots\dots(1)$$

Where $a_0, a_1, b_1, a_2, b_2, \dots$ are all constants called as the **co-efficients** of the series such that **every term of the series** is **periodic of period 2π** (if the co-efficients are such that the series converges then its sum will be a function of period 2π).

Let $f(x)$ be a given function of period 2π such that it can be represented by the infinite series given by eqn(1) i.e eqn(1) converges and moreover has the sum $f(x)$ then using the equality sign we write

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_n \cos(nx) + b_n \sin(nx)\} \quad \dots\dots\dots(2)$$

Where eqn(2) is called as the **Infinite Trigonometric Fourier Series** expansion of $f(x)$ (or) simply called as the **Fourier Series of $f(x)$** .

The co-efficients of eqn(2) are called as **Fourier co-efficients** of $f(x)$ which is given by Euler formulas as

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \quad \dots\dots\dots(3a)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx, \quad n = 1, 2, 3, \dots \quad \dots\dots\dots(3b)$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx, \quad n = 1, 2, 3, \dots \quad \dots\dots\dots(3c)$$

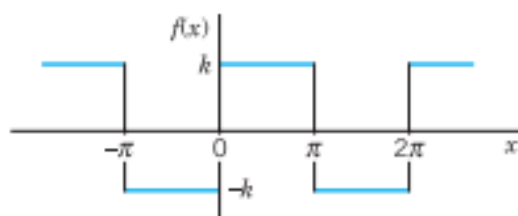
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Note: Leonard Euler(1707-1783) was an enormously great creative Swiss mathematician. He made fundamental contributions to almost all branches of mathematics and its applications to Physics. His important books on algebra and calculus contains numerous basic results of his own research.

Basic Example:

- 1) Find the Fourier co-efficients of the periodic rectangular wave function $f(x)$ as shown in

the figure below $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$ and $f(x + 2\pi) = f(x)$



(functions of this kind act as external forces acting on mechanical systems, electromotive forces in electric circuits, etc... The value of $f(x)$ at a single point does not affect the integral. Hence we can leave $f(x)$ undefined at $x = 0$ & $x = \pm \pi$)

Solt: By data , $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$ (1)

$f(x + 2\pi) = f(x)$ comparing with $f(x + p) = f(x)$ we have $p = 2\pi$

i.e the given function $f(x)$ by eqn(1) is periodic of period $p = 2\pi$ (2)

we use Euler's formulas to find the Fourier co-efficients a_0, a_n & b_n

wkt, $a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$, using eqn (1) we get

$$\Rightarrow a_0 = \frac{1}{2\pi} \left[\int_{-\pi}^0 f(x) dx + \int_0^{\pi} f(x) dx \right]$$

$$\Rightarrow a_0 = \frac{1}{2\pi} \left[\int_{-\pi}^0 (-k) dx + \int_0^{\pi} (k) dx \right]$$

$$\Rightarrow a_0 = \frac{1}{2\pi} [\{(-k) x\}_{-\pi}^0 + \{(k) x\}_0^{\pi}]$$

$$\Rightarrow a_0 = \frac{1}{2\pi} [(-k)\{0 - (-\pi)\} + (k)\{\pi - 0\}]$$

$$\Rightarrow a_0 = \frac{1}{2\pi} [-k\pi + k\pi] \quad \therefore \quad a_0 = 0 \quad \text{.....(3)}$$

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Wkt, $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$, $n = 1, 2, 3, \dots$, using eqn. (1) we get

$$\Rightarrow a_n = \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) \cos(nx) dx + \int_0^{\pi} f(x) \cos(nx) dx \right]$$

$$\Rightarrow a_n = \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \cos(nx) dx + \int_0^{\pi} (k) \cos(nx) dx \right]$$

$$\Rightarrow a_n = \frac{1}{\pi} \left[(-k) \left\{ \frac{\sin(nx)}{n} \right\}_{-\pi}^0 + (k) \left\{ \frac{\sin(nx)}{n} \right\}_0^{\pi} \right]$$

$$\Rightarrow a_n = \frac{1}{\pi} \left[(-1) \frac{k}{n} \{\sin(nx)\}_{-\pi}^0 + \frac{k}{n} \{\sin(nx)\}_0^{\pi} \right]$$

$$\Rightarrow a_n = \frac{1}{\pi} * \left(\frac{k}{n} \right) [-\{\sin(nx)\}_{-\pi}^0 + \{\sin(nx)\}_0^{\pi}]$$

$$\Rightarrow a_n = \frac{1}{\pi} * \left(\frac{k}{n} \right) [-\{0 - 0\} + \{0 - 0\}] \because \sin(0) = \sin(\pm n\pi) = 0, n = 1, 2, 3 \dots$$

$$\therefore a_n = 0 \dots\dots\dots(4)$$

Wkt, $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$, $n = 1, 2, 3, \dots$ using eqn. (1) we get

$$\Rightarrow b_n = \frac{1}{\pi} \left[\int_{-\pi}^0 f(x) \sin(nx) dx + \int_0^{\pi} f(x) \sin(nx) dx \right]$$

$$\Rightarrow b_n = \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \sin(nx) dx + \int_0^{\pi} (k) \sin(nx) dx \right]$$

$$\Rightarrow b_n = \frac{1}{\pi} \left[(-k) \left\{ \frac{-\cos(nx)}{n} \right\}_{-\pi}^0 + (k) \left\{ \frac{-\cos(nx)}{n} \right\}_0^{\pi} \right]$$

$$\Rightarrow b_n = \frac{1}{\pi} \left[\left(\frac{k}{n} \right) \{\cos(nx)\}_{-\pi}^0 - \left(\frac{k}{n} \right) \{\cos(nx)\}_0^{\pi} \right]$$

$$\Rightarrow b_n = \frac{1}{\pi} * \left(\frac{k}{n} \right) [\{1 - \cos(n\pi)\} - \{\cos(n\pi) - 1\}] \because \cos(-n\pi) = \cos(n\pi)$$

$$\Rightarrow b_n = \frac{1}{\pi} * \left(\frac{k}{n} \right) [\{1 - \cos(n\pi)\} + \{1 - \cos(n\pi)\}]$$

$$\Rightarrow b_n = \frac{2}{\pi} * \left(\frac{k}{n} \right) [1 - \cos(n\pi)] \dots\dots\dots(5a)$$

Wkt $\cos(\pi) = -1, \cos(2\pi) = 1, \cos(3\pi) = -1, \cos(4\pi) = 1, \dots\dots\dots$

Thus $\cos(n\pi) = \begin{cases} -1, & \text{when } n \text{ is odd} \\ 1, & \text{when } n \text{ is even} \end{cases} \therefore 1 - \cos(n\pi) = \begin{cases} 2, & \text{when } n \text{ is odd} \\ 0, & \text{when } n \text{ is even} \end{cases}$

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Thus eqn. (5a) can be written as

$$\Rightarrow b_1 = \frac{4k}{\pi}, b_2 = 0, \Rightarrow b_3 = \frac{4k}{3\pi}, b_4 = 0, b_5 = \frac{4k}{5\pi}, \dots \dots \dots (5b)$$

Thus substituting eqns. (3), (4), (5b) in Fourier series expansion given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_n \cos(nx) + b_n \sin(nx)\}$$

$$\Rightarrow f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$\Rightarrow f(x) = 0 + 0 + \{b_1 \sin(x) + b_2 \sin(2x) + b_3 \sin(3x) + b_4 \sin(4x) + b_5 \sin(5x) + \dots\}$$

$$\Rightarrow f(x) = \frac{4k}{\pi} \left[\sin x + \frac{1}{3} \sin(3x) + \frac{1}{5} \sin(5x) + \dots \right] \dots \dots \dots (6)$$

gives the Fourier series expansion of the function $f(x)$ given by eqn. (1)

From the above equation the partial sums are:

$$S_1 = \frac{4k}{\pi} \sin x, S_2 = \frac{4k}{\pi} [\sin x + \sin 3x], \dots \dots \dots (7)$$

As per the figure given in the data indicates that the series is convergent and has the sum (x) , which is the given function.

Also, we observe that at the points $x = 0$ & at $x = \pi$ which are the points of discontinuity of the given function $f(x)$ where all the partial sums have the value zero which is the arithmetic mean of the values $-k$ & k of our function at these points.

Note: Let us assume $x = \pi/2$ in eqn. (6), we get

$$\Rightarrow f(\pi/2) = \frac{4k}{\pi} \left[\sin\left(\frac{\pi}{2}\right) + \frac{1}{3} \sin\left(3\frac{\pi}{2}\right) + \frac{1}{5} \sin\left(5\frac{\pi}{2}\right) + \dots \right]$$

$$\Rightarrow k = \frac{4k}{\pi} \left[1 - \frac{1}{3} + \frac{1}{5} + \dots \right]$$

$$\therefore \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \dots \dots \dots (8)$$

Eqn. (8) is a famous result obtained by *Leibnitz* in 1673 from geometric considerations. It illustrates that the “values of various series with constant terms can be obtained by evaluating Fourier series at specific points”.

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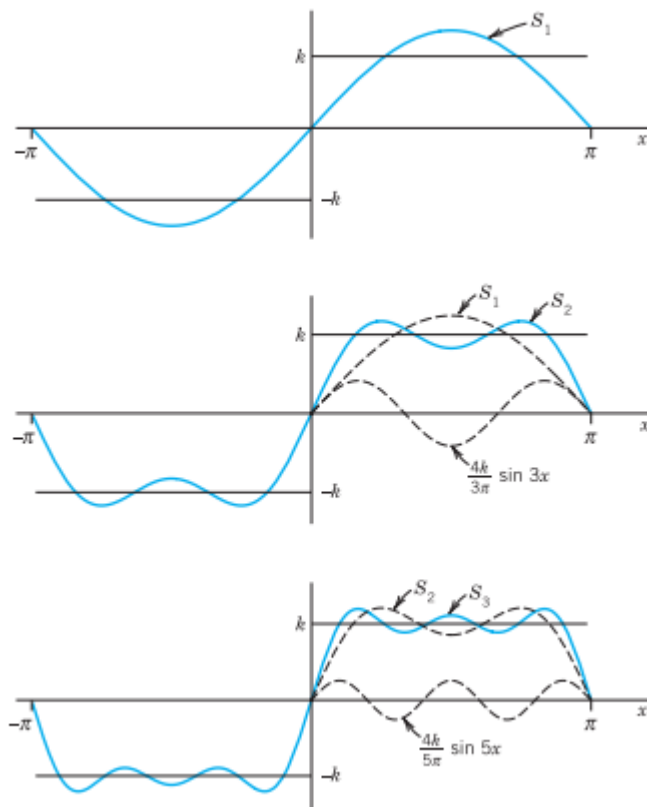


Figure: Graphs of first three partial sums.

Conditions for the existence of Fourier series:

Only those functions which satisfy a set of conditions over a period can be expressed as an infinite Fourier series i.e as an infinite series of trigonometric sine and cosine functions

($\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$) and the conditions are:

- 1) $f(x)$ is **periodic** of period 2π .
- 2) $f(x)$ is a **single valued function** and **finite** over the interval $(c, c + 2\pi)$.

$$\int_{x=c}^{c+2l} |f(x)| dx \neq \pm\infty$$

- 3) $f(x)$ has finite number of discontinuities over the interval $(c, c + 2\pi)$.

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- 4) $f(x)$ has atmost finite number of maxima and minima over the interval $(c, c + 2\pi)$.

Above conditions are called as **Dirichlet's** conditions.

Thus if the function $f(x)$ is defined over the interval $(c, c + 2\pi)$ and satisfies Dirichlet's conditions then the infinite trigonometric series

$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$ is called as **Fourier series** of $f(x)$ over the interval $(c, c + 2\pi)$.

Note:

- 1) If $f(x)$ is discontinuous at $x = x_0 \in (c, c + 2\pi)$ then the infinite trigonometric Fourier series converges to $\frac{1}{2} [f(x^+) + f(x^-)]$ where,
 $f(x^+) = \lim_{h \rightarrow 0} f(x + h)$, the right hand limit
 $f(x^-) = \lim_{h \rightarrow 0} f(x - h)$, the left hand limit
 $h > 0$
- 2) If $f(x)$ is discontinuous at end points of the interval $(c, c + 2\pi)$ i.e either at $x = c$ or at $x = c + 2\pi$ then $f(x)$ converges to $\frac{1}{2} \{f(c) + f(c + 2\pi)\}$
- 3) $f(x) = \tan(x)$ is an example of a function which cannot be expressed as Fourier series as it has infinite number of infinite discontinuity.

Pre-requisites :

- 1) $\cos\left((2n + 1)\frac{\pi}{2}\right) = 0, \quad n = 0, 1, 2, 3, 4, \dots$
 $\therefore \cos\left(\frac{\pi}{2}\right) = \cos\left(3\frac{\pi}{2}\right) = \cos\left(5\frac{\pi}{2}\right) = \dots = 0$
- 2) $\sin(n\pi) = 0, \quad n = 0, 1, 2, 3, 4, \dots \quad \therefore \sin(0) = \sin(\pi) = \sin(2\pi) = \dots = 0$
- 3) $\cos(n\pi) = (-1)^n, \quad n = 0, 1, 2, 3, 4, \dots \quad (or)$
 $\cos(n\pi) = \begin{cases} +1, & n = 0, 2, 4, 6, \dots \quad (n = \text{even}) \\ -1, & n = 1, 3, 5, \dots \quad (n = \text{odd}) \end{cases}$

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$$\cos(0) = \cos(2\pi) = \cos(4\pi) = \dots = 1 \quad \& \quad \cos(\pi) = \cos(3\pi) = \cos(5\pi) = \dots = -1$$

$$4) \quad \sin\left((2n+1)\frac{\pi}{2}\right) = (-1)^n, \quad n = 0, 1, 2, 3, 4, \dots$$

$$(or) \quad \sin\left((2n+1)\frac{\pi}{2}\right) = \begin{cases} +1, & n = 0, 2, 4, \dots \quad (n = \text{even}) \\ -1, & n = 1, 3, 5, \dots \quad (n = \text{odd}) \end{cases}$$

$$i.e \quad \sin\left(\frac{\pi}{2}\right) = \sin\left(5\frac{\pi}{2}\right) = \sin\left(9\frac{\pi}{2}\right) = \dots = 1$$

$$\& \quad \sin\left(3\frac{\pi}{2}\right) = \sin\left(7\frac{\pi}{2}\right) = \sin\left(11\frac{\pi}{2}\right) = \dots = -1$$

$$5) \quad \int_{x=-a}^a f(x) dx = \begin{cases} 2 \int_{x=0}^a f(x) dx, & f(-x) = f(x) \text{ i.e } f(x) \text{ is an even function} \\ 0, & f(-x) = -f(x) \text{ i.e } f(x) \text{ is an odd function} \end{cases}$$

$$6) \quad \int_{x=0}^{2a} f(x) dx = \begin{cases} 2 \int_{x=0}^a f(x) dx, & f(2a-x) = f(x) \text{ i.e } f(x) \text{ is an even function} \\ 0, & f(2a-x) = -f(x) \text{ i.e } f(x) \text{ is an odd function} \end{cases}$$

7) **Bernoulli's rule** of integration by parts:

$$\int uv = u_0 v_1 - u_1 v_2 + u_2 v_3 - u_3 v_4 + \dots$$

$$(or) \quad \int uv = u_0 v_1 - u' v_2 + u'' v_3 - u''' v_4 + \dots$$

Where $u_0, u_1, u_2, u_3, \dots$ i.e all u 's represent successive derivatives of u

$v_0, v_1, v_2, v_3, \dots$ i.e all v 's represent successive integrals of v

$$8) \quad \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2+b^2} \{a \sin bx - b \cos bx\}, \quad a \& b \text{ are constants}$$

$$9) \quad \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2+b^2} \{a \cos bx + b \sin bx\}, \quad a \& b \text{ are constants}$$

$$10) \quad \text{length of an open interval } (a, b) = b - a, \quad a < b$$

$$11) \quad \cos(x) \text{ is an } \mathbf{even} \text{ function whereas } \sin(x) \text{ is an } \mathbf{odd} \text{ function}$$

$$12) \quad \text{Product of two even functions are always } \mathbf{even}. \text{ Similarly, Product of two odd functions are always } \mathbf{even}.$$

$$13) \quad \text{The product of an even function with an odd function is always an } \mathbf{odd} \text{ function.}$$

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$$14) \int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx = \begin{cases} 0, & n \neq m \\ \pi, & n = m > 0 \end{cases}$$

$$15) \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx = \begin{cases} 0, & n \neq m \\ \pi, & n = m > 0 \end{cases}$$

Arbitrary period, Odd & Even Functions :

Fourier series of arbitrary period:

Let the function $f(x)$ be defined over the general interval $(c, c + 2l)$ whose period length is $2l$ then $f(x)$ after satisfying Dirichlet's conditions in $(c, c + 2l)$ we have infinite trigonometric Fourier series (or simply Fourier series) of $f(x)$ given as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \dots\dots\dots(1)$$

Where a_0, a_n & b_n are constants called as **Fourier co-efficients** which are given by **Euler's** formulas as

$$a_0 = \frac{1}{l} \int_{x=c}^{c+2l} f(x) dx \dots\dots\dots(2)$$

$$a_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx \dots\dots\dots(3)$$

$$b_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx \dots\dots\dots(4)$$

Types of intervals:

There are four types: $(0, 2\pi), (-\pi, \pi), (0, 2l)$ & $(-l, l)$

Case(1): $(c, c + 2l) = (0, 2\pi) \therefore c = 0 \text{ \& } l = \pi$

Therefore Fourier series and Euler's formulas given by eqns. (1), (2), (3), (4) reduces to

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx) \dots\dots\dots(5)$$

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$$a_0 = \frac{1}{\pi} \int_{x=0}^{2\pi} f(x) dx \quad \dots\dots\dots(6)$$

$$a_n = \frac{1}{\pi} \int_{x=0}^{2\pi} f(x) \cos(nx) dx \quad \dots\dots\dots(7)$$

$$b_n = \frac{1}{\pi} \int_{x=0}^{2\pi} f(x) \sin(nx) dx \quad \dots\dots\dots(8)$$

Case(2): $(c, c + 2l) = (-\pi, \pi) \quad \therefore \quad c = -l \text{ \& \; } l = \pi$

Therefore Fourier series and Euler's formulas given by eqns. (1), (2), (3), (4) reduces to

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx) \quad \dots\dots\dots(9)$$

$$a_0 = \frac{1}{\pi} \int_{x=-\pi}^{\pi} f(x) dx \quad \dots\dots\dots(10)$$

$$a_n = \frac{1}{\pi} \int_{x=-\pi}^{\pi} f(x) \cos(nx) dx \quad \dots\dots\dots(11)$$

$$b_n = \frac{1}{\pi} \int_{x=-\pi}^{\pi} f(x) \sin(nx) dx \quad \dots\dots\dots(12)$$

Case(3): $(c, c + 2l) = (0, 2l) \quad \therefore \quad c = 0$

Therefore Fourier series and Euler's formulas given by eqns. (1), (2), (3), (4) reduces to

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \quad \dots\dots\dots(13)$$

$$a_0 = \frac{1}{l} \int_{x=0}^{2l} f(x) dx \quad \dots\dots\dots(14)$$

$$a_n = \frac{1}{l} \int_{x=0}^{2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx \quad \dots\dots\dots(15)$$



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$$b_n = \frac{1}{l} \int_{x=0}^{2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx \quad \dots\dots\dots(16)$$

Case(4): $(c, c + 2l) = (-l, l) \quad \therefore c = -l$

Therefore Fourier series and Euler's formulas given by eqns. (1), (2), (3), (4) reduces to

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \quad \dots\dots\dots(17)$$

$$a_0 = \frac{1}{l} \int_{x=-l}^l f(x) dx \quad \dots\dots\dots(18)$$

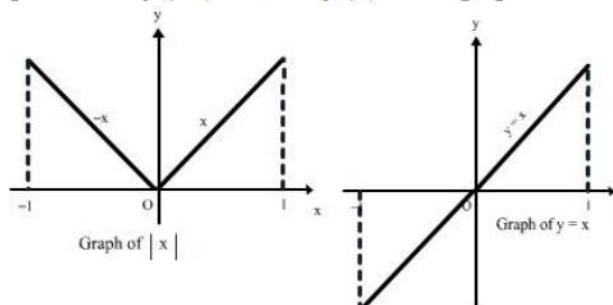
$$a_n = \frac{1}{l} \int_{x=-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \quad \dots\dots\dots(19)$$

$$b_n = \frac{1}{l} \int_{x=-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx \quad \dots\dots\dots(20)$$

Even & Odd Nature of the Function (x) :

Generally any function $f(x)$ is said to be even when $f(-x) = f(x)$ which is always symmetrical about $y - axis$ (i.e $y - axis$ acts like a mirror) and odd when $f(-x) = -f(x)$, which is always symmetrical about *origin*.

For example, the function $f(x) = |x|$ in $[-1,1]$ is even as $f(-x) = |-x| = |x| = f(x)$ and the function $f(x) = x$ in $[-1,1]$ is odd as $f(-x) = -x = -f(x)$. The graphs of these functions are shown below :



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In Fourier series even and odd nature of the function $f(x)$ will depend upon the interval $(0, 2l)$ (or) $(0, 2\pi)$ & $(-l, l)$ (or) $(-\pi, \pi)$ (refer page no:12, pts (5) & (6))

Even Function:

Case(1) : Let $f(x)$ be an **even** function in the interval $(c, c + 2l) = (0, 2\pi)$ then

$$a_0 = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) dx \quad (\text{from pt (6) page (12), case(1)-page(14) })$$

$$a_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \cos(nx) dx \quad (\text{from pt (6) & (12) pg (12), case(1)-pg(14) })$$

$\because f(x) \rightarrow \text{even function} \text{ \& } \cos(nx) \rightarrow \text{even function}$

$$b_n = 0 \quad (\text{from pt (6) & (13) pg (12, 13), case(1)-pg(14) })$$

$\because f(x) \rightarrow \text{even function} \text{ \& } \sin(nx) \rightarrow \text{odd function}$

Case(2) : Let $f(x)$ be an **even** function in the interval $(c, c + 2l) = (0, 2l)$ then

$$a_0 = \frac{2}{l} \int_{x=0}^l f(x) dx \quad (\text{from pt (6) page (12), case(3)-page(14) })$$

$$a_n = \frac{2}{l} \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \quad (\text{from pt (6) & (12) pg (12), case(3)-pg(14)})$$

$\because f(x) \rightarrow \text{even function} \text{ \& } \cos\left(\frac{n\pi x}{l}\right) \rightarrow \text{even function}$

$$b_n = 0 \quad (\text{from pt (6) & (13) pg (12, 13), case(3)-pg(14)})$$

$\because f(x) \rightarrow \text{even function} \text{ \& } \sin(nx) \rightarrow \text{odd function}$

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Case(3) : Let $f(x)$ be an **even** function in the interval $(c, c + 2l) = (-\pi, \pi)$ then

$$a_0 = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) dx \quad (\text{from pt (5) page (12), case(2)-page(14)})$$

$$a_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \cos(nx) dx \quad (\text{from pt (5) \& (12) pg (12), case(2)-pg(14)})$$

$\because f(x) \rightarrow \text{even function} \ \& \ \cos(nx) \rightarrow \text{even function}$

$$b_n = 0 \quad (\text{from pt (5) \& (13) pg (12, 13), case(2)-pg(14)})$$

$\because f(x) \rightarrow \text{even function} \ \& \ \sin(nx) \rightarrow \text{odd function}$

Case(4) : Let $f(x)$ be an **even** function in the interval $(c, c + 2l) = (-l, l)$ then

$$a_0 = \frac{2}{l} \int_{x=0}^l f(x) dx \quad (\text{from pt (5) page (12), case(4)-page(15) })$$

$$a_n = \frac{2}{l} \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \quad (\text{from pt (5) \& (12) pg (12), case(4)-pg(15)})$$

$\because f(x) \rightarrow \text{even function} \ \& \ \cos\left(\frac{n\pi x}{l}\right) \rightarrow \text{even function}$

$$b_n = 0 \quad (\text{from pt (5) \& (13) pg (12, 13), case(4)-pg(15)})$$

$\because f(x) \rightarrow \text{even function} \ \& \ \sin(nx) \rightarrow \text{odd function}$

Odd Function:

Case(1) : Let $f(x)$ be an **odd** function in the interval $(c, c + 2l) = (0, 2\pi)$ then

$$a_0 = 0 \quad (\text{from pt (6) page (12), case(1)-page(14) })$$

$$a_n = 0 \quad (\text{from pt (6) \& (13) pg (12), case(1)-pg(14) })$$

$\because f(x) \rightarrow \text{odd function} \ \& \ \cos(nx) \rightarrow \text{even function}$

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$$b_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin(nx) dx \text{ (from pt (6) \& (12) pg (12, 13), case(1)-pg(14))}$$

$\because f(x) \rightarrow \text{odd function} \ \& \ \sin(nx) \rightarrow \text{odd function}$

Case(2) : Let $f(x)$ be an **odd** function in the interval $(c, c + 2l) = (0, 2l)$ then

$$a_0 = 0 \quad \text{(from pt (6) page (12), case(3)-page(14))}$$

$$a_n = 0 \quad \text{(from pt (6) \& (13) pg (12), case(3)-pg(14))}$$

$\because f(x) \rightarrow \text{odd function} \ \& \ \cos(nx) \rightarrow \text{even function}$

$$b_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin\left(\frac{n\pi x}{l}\right) dx \text{ (from pt (6) \& (12) pg (12, 13), case(3)-pg(14))}$$

$\because f(x) \rightarrow \text{odd function} \ \& \ \sin(nx) \rightarrow \text{odd function}$

Case(3) : Let $f(x)$ be an **odd** function in the interval $(c, c + 2l) = (-\pi, \pi)$ then

$$a_0 = 0 \quad \text{(from pt (5) page (12), case(2)-page(14))}$$

$$a_n = 0 \quad \text{(from pt (5) \& (13) pg (12), case(2)-pg(14))}$$

$\because f(x) \rightarrow \text{Odd function} \ \& \ \cos(nx) \rightarrow \text{even function}$

$$b_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin(nx) dx \text{ (from pt (5) \& (12) pg (12, 13), case(2)-pg(14))}$$

$\because f(x) \rightarrow \text{odd function} \ \& \ \sin(nx) \rightarrow \text{odd function}$

Case(4) : Let $f(x)$ be an **odd** function in the interval $(c, c + 2l) = (-l, l)$ then

$$a_0 = 0 \quad \text{(from pt (5) page (12), case(4)-page(15))}$$

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$$a_n = 0$$

(from pt (5) & (13) pg (12), case(4)-pg(15))

$$\because f(x) \rightarrow \text{odd function} \quad \& \quad \cos\left(\frac{n\pi x}{l}\right) \rightarrow \text{even function}$$

$$b_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin\left(\frac{n\pi x}{l}\right) dx \quad (\text{from pt (5) \& (12) pg (12, 13), case(4)-pg(15)})$$

$$\because f(x) \rightarrow \text{odd function} \quad \& \quad \sin\left(\frac{n\pi x}{l}\right) \rightarrow \text{odd function}$$

Above obtained results can be formulated in the table as shown below

Interval	Type of fn $f(x)$	a_0	a_n	b_n
$(0, 2\pi)$	Even	$\frac{2}{\pi} \int_{x=0}^{\pi} f(x) dx$	$\frac{2}{\pi} \int_{x=0}^{\pi} f(x) \cos(nx) dx$	0
$(0, 2l)$	Even	$\frac{2}{l} \int_{x=0}^l f(x) dx$	$\frac{2}{l} \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$	0
$(-\pi, \pi)$	Even	$\frac{2}{\pi} \int_{x=0}^{\pi} f(x) dx$	$\frac{2}{\pi} \int_{x=0}^{\pi} f(x) \cos(nx) dx$	0
$(-l, l)$	Even	$\frac{2}{l} \int_{x=0}^l f(x) dx$	$\frac{2}{l} \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$	0
$(0, 2\pi)$	Odd	0	0	$\frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin(nx) dx$
$(0, 2l)$	Odd	0	0	$\frac{2}{l} \int_{x=0}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$
$(-\pi, \pi)$	Odd	0	0	$\frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin(nx) dx$
$(-l, l)$	Odd	0	0	$\frac{2}{l} \int_{x=0}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$

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Problems on Fourier Series Expansion:

Interval $(0, 2\pi)$ & $(-\pi, \pi)$

Solve the Following:

- 1) Express $f(x) = (\pi - x)^2$ as a Fourier series of period 2π in the interval $0 < x < 2\pi$. Hence deduce the sum of the series $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

Solt: By data $f(x) = (\pi - x)^2$ (1) where $0 < x < 2\pi$ (or) $x \in (0, 2\pi)$ (2)

Comparing $(0, 2\pi)$ with $(c, c + 2l)$ we get $c = 0$ & $l = \pi$ (3)

Wkt the Fourier series expansion $f(x)$ over the interval $(c, c + 2l)$ is given as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \text{(4)}$$

where a_0 , a_n & b_n are the Fourier co-efficients which are given by Euler's formulae as

$$a_0 = \frac{1}{l} \int_{x=c}^{c+2l} f(x) dx \text{(5)}$$

$$a_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx \text{(6)}$$

$$b_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx \text{(7)}$$

Finding the nature of the given function $f(x) = (\pi - x)^2$, $x \in (0, 2\pi)$

Wkt

$$\int_{x=0}^{2a} f(x) dx = \begin{cases} 2 \int_{x=0}^a f(x) dx, & f(2a - x) = f(x) \text{ i.e } f(x) \text{ is an even function} \\ 0, & f(2a - x) = -f(x) \text{ i.e } f(x) \text{ is an odd function} \end{cases}$$

$$f(2\pi - x) = (\pi - \{2\pi - x\})^2$$

$$\Rightarrow f(2\pi - x) = (\pi - 2\pi + x)^2$$

$$\Rightarrow f(2\pi - x) = (-\pi + x)^2 = \{-(\pi - x)\}^2$$

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$$\Rightarrow f(2\pi - x) = (\pi - x)^2 = f(x)$$

$$\therefore f(x) = (\pi - x)^2 \text{ is an even function} \quad \therefore b_n = 0 \quad \dots\dots\dots(8)$$

Using eqns. (1),(2),(3) & (8) in eqns. (5), (6), (7) we get

$$a_0 = \frac{1}{\pi} \int_{x=0}^{2\pi} f(x) dx \quad \Rightarrow a_0 = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) dx \quad \because f(x) \text{ is an even fn}$$

$$\Rightarrow a_0 = \frac{2}{\pi} \int_{x=0}^{\pi} (\pi - x)^2 dx \quad \Rightarrow a_0 = \frac{2}{\pi} \left\{ \frac{(\pi - x)^3}{(-3)} \right\}_0^{\pi}$$

$$\Rightarrow a_0 = \frac{2}{-3\pi} \{(\pi - x)^3\}_0^{\pi} \quad \Rightarrow a_0 = \frac{-2}{3\pi} \{(\pi - \pi)^3 - (\pi - 0)^3\}$$

$$\Rightarrow a_0 = \frac{-2}{3\pi} \{0 - \pi^3\} \quad \Rightarrow a_0 = \frac{-2}{3\pi} \{-\pi^3\}$$

$$a_0 = \frac{2\pi^2}{3} \quad \dots\dots\dots \rightarrow (9a) \quad (\text{or}) \quad \frac{a_0}{2} = \frac{\pi^2}{3} \quad \dots\dots\dots \rightarrow (9b)$$

Wkt from eqn.(6)

$$a_n = \frac{1}{\pi} \int_{x=0}^{2\pi} f(x) \cos\left(\frac{n\pi x}{\pi}\right) dx$$

$$\Rightarrow a_n = \frac{2}{\pi} \int_{x=0}^{\pi} (\pi - x)^2 \cos(nx) dx \quad \because f(x) \text{ \& } \cos(nx) \text{ are both even fns} \quad \dots\dots \rightarrow (10a)$$

By using Bernoulli's rule of integration by parts, where

$$\int uv = u_0 v_1 - u_1 v_2 + u_2 v_3 - u_3 v_4 + \dots$$

$u = u_0 = (\pi - x)^2$	$v = \cos(nx)$
$u_1 = u' = 2(\pi - x)^1 * (-1)$	$\therefore v_1 = \int \cos(nx) dx \therefore v_1 = \frac{\sin(nx)}{n}$
$u_2 = u'' = (-2)(-1) \therefore u_2 = u'' = 2$	$\therefore v_2 = \int \frac{\sin(nx)}{n} dx \therefore v_2 = \frac{-\cos(nx)}{n^2}$
$u_3 = u''' = 0$	$\therefore v_3 = \int \frac{-\cos(nx)}{n^2} dx \therefore v_3 = \frac{-\sin(nx)}{n^3}$

Substituting the values obtained from the table in the above Bernoulli's equation we get

$$\begin{aligned} \int_{x=0}^{\pi} (\pi - x)^2 \cos(nx) dx &= \left\{ (\pi - x)^2 * \left(\frac{\sin(nx)}{n} \right) \right\}_0^{\pi} - \left\{ -2(\pi - x)^1 * \left(\frac{-\cos(nx)}{n^2} \right) \right\}_0^{\pi} \\ &\quad + \left\{ (2) * \left(\frac{-\sin(nx)}{n^3} \right) \right\}_0^{\pi} \end{aligned}$$

$$\because \sin(0) = \sin(n\pi) = 0, \quad n = 1, 2, 3, 4, \dots$$

$$\Rightarrow \int_{x=0}^{\pi} (\pi - x)^2 \cos(nx) dx = \{0 - 0\} - \left\{ \frac{2}{n^2} \right\} \{(\pi - x) * \cos(nx)\}_0^{\pi} + \left\{ \frac{-2}{n^3} \right\} \{0 - 0\}$$

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$$\Rightarrow \int_{x=0}^{\pi} (\pi - x)^2 \cos(nx) dx = - \left\{ \frac{2}{n^2} \right\} [\{ (\pi - \pi) * \cos(n\pi) \} - \{ (\pi - 0) * \cos(0) \}]$$

$$\Rightarrow \int_{x=0}^{\pi} (\pi - x)^2 \cos(nx) dx = - \left\{ \frac{2}{n^2} \right\} \{ 0 - (\pi) \} \quad \because \cos(0) = 1$$

$$\Rightarrow \int_{x=0}^{\pi} (\pi - x)^2 \cos(nx) dx = - \left\{ \frac{2}{n^2} \right\} \{ -\pi \}, n = 1, 2, 3, 4, \dots$$

$$\Rightarrow \int_{x=0}^{\pi} (\pi - x)^2 \cos(nx) dx = \frac{2\pi}{n^2}, n = 1, 2, 3, 4, \dots \quad \text{-----} \rightarrow (10b)$$

Substituting eqn. (10b) in the RHS of eqn. (10a) we get

$$\Rightarrow a_n = \frac{2}{\pi} * \frac{2\pi}{n^2}, n = 1, 2, 3, 4, \dots$$

$$\Rightarrow a_n = \frac{4}{n^2}, n = 1, 2, 3, 4, \dots \text{ \& } n \neq 0 \quad \text{-----} \rightarrow (10c)$$

Substituting for a_0, a_n & b_n from eqn.(9b), (5), (10c) & (8) in RHS of eqn. (4) we get

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} * \cos\left(\frac{n\pi x}{\pi}\right) + 0$$

$$f(x) = \frac{\pi^2}{3} + 4 * \sum_{n=1}^{\infty} \frac{\cos(nx)}{n^2} \quad (\text{or})$$

$$(\pi - x)^2 = \frac{\pi^2}{3} + 4 * \sum_{n=1}^{\infty} \frac{\cos(nx)}{n^2}$$

-----> (11)

Thus eqn. (11) gives the Fourier series expansion of $f(x) = (\pi - x)^2$ in the interval $x \in (0, 2\pi)$

To deduce the sum of the series $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

put $x = 0$ on both the sides of eqn. (11) but $x = 0 \notin (0, 2\pi)$

i.e $x = 0$ is an end point at which $f(x) = (\pi - x)^2$ is discontinuous

$$\therefore f(x) \rightarrow \frac{1}{2} [f(0) + f(2\pi)] \quad \text{-----} > (12a)$$

To find $f(0)$ put $x = 0$ & to find $f(2\pi)$ put $x = 2\pi$ in $f(x) = (\pi - x)^2$ we get

$f(0) = (\pi - 0)^2 \therefore f(0) = \pi^2 \quad f(2\pi) = (\pi - 2\pi)^2 \Rightarrow f(2\pi) = (-\pi)^2 \therefore f(2\pi) = \pi^2$

$\therefore$ LHS of eqn. (11) is : $LHS = \frac{1}{2} [\pi^2 + \pi^2] \therefore LHS = \pi^2$ in eqn.(11) & put $x = 0$

$$\pi^2 = \frac{\pi^2}{3} + 4 * \sum_{n=1}^{\infty} \frac{\cos(0)}{n^2} \quad \Rightarrow \quad \pi^2 - \frac{\pi^2}{3} = 4 * \sum_{n=1}^{\infty} \frac{1}{n^2}$$

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$$\Rightarrow \frac{2\pi^2}{3} = 4 * \sum_{n=1}^{\infty} \frac{1}{n^2} \quad \Rightarrow \frac{2\pi^2}{4*3} = \sum_{n=1}^{\infty} \frac{1}{n^2} \quad \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \quad (or) \quad 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$

- 2) Express $f(x) = (\pi - x)$ as a Fourier series of period 2π in the interval $0 < x < 2\pi$. Hence deduce the sum of the series $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

Solt: By data $f(x) = (\pi - x)$ -----→(1) where $x \in (0, 2\pi)$ -----(2)

Comparing the interval $(c, c + 2l)$ with $(0, 2\pi)$ we get

$$\boxed{c = 0 \text{ \& } l = \pi} \quad \text{-----→(3)}$$

To find the nature of the given function $f(x) = (\pi - x)$ where $x \in (0, 2\pi)$

$$f(2\pi - x) = \pi - (2\pi - x) \Rightarrow f(2\pi - x) = \pi - 2\pi + x \Rightarrow f(2\pi - x) = -\pi + x$$

$$\therefore f(2\pi - x) = -(\pi - x) = -f(x) \quad \therefore f(2\pi - x) = -f(x)$$

$\therefore f(x) = (\pi - x)$ is an **odd** function in the interval $x \in (0, 2\pi)$

$$\boxed{\therefore a_0 = 0} \quad \text{-----→(4)}$$

$$\boxed{\therefore a_n = 0} \quad \text{-----→(5)}$$

Wkt $b_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$, $c = 0$ & $l = \pi$

$$\Rightarrow b_n = \frac{1}{\pi} \int_{x=0}^{2\pi} f(x) \sin\left(\frac{n\pi x}{\pi}\right) dx, \quad f(x) \text{ \& } \text{sine fn. both are odd}$$

$$\Rightarrow b_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin(nx) dx$$

$$\Rightarrow b_n = \frac{2}{\pi} \int_{x=0}^{\pi} (\pi - x) \sin(nx) dx \quad \text{-----→(6a)}$$

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By applying Bernoulli's rule of integration by parts,

$$\int uv = u_0 v_1 - u_1 v_2 + u_2 v_3 - u_3 v_4 + \dots$$

$u = u_0 = (\pi - x)$	$v = \sin(nx)$
$u_1 = u' = -1$	$\therefore v_1 = \int \sin(nx) dx \therefore v_1 = \frac{-\cos(nx)}{n}$
$u_2 = u'' = 0$	$\therefore v_2 = \int \frac{-\cos(nx)}{n} dx \therefore v_2 = \frac{-\sin(nx)}{n^2}$

Substituting above obtained values from the table in the RHS of eqn. (6a) we get

$$\Rightarrow b_n = \frac{2}{\pi} \left(\left\{ (\pi - x) * \left(\frac{-\cos(nx)}{n} \right) \right\}_0^\pi - \left\{ (-1) * \left(\frac{-\sin(nx)}{n^2} \right) \right\}_0^\pi + 0 \right)$$

$$\Rightarrow b_n = \frac{2}{\pi} \left(\frac{-1}{n} \{ (\pi - x) * (\cos(nx)) \}_0^\pi - \{ 0 - 0 \} \right) \because \sin(0) = \sin(\pi) = \sin(2\pi) = \dots = 0$$

$$\Rightarrow b_n = \frac{2}{\pi} * \left(\frac{-1}{n} \right) * \{ (\pi - x) * \cos(nx) \}_0^\pi$$

$$\Rightarrow b_n = \frac{-2}{\pi} * \left(\frac{1}{n} \right) * \{ [(\pi - \pi) * \cos(n\pi)] - [(\pi - 0) * \cos(0)] \}$$

$$\Rightarrow b_n = \frac{-2}{\pi} * \left(\frac{1}{n} \right) * \{ 0 - \pi \} \because \cos(0) = 1 \quad \Rightarrow b_n = \frac{-2}{\pi} * \left(\frac{1}{n} \right) * (-\pi)$$

$$\Rightarrow b_n = 2 * \left(\frac{1}{n} \right) \Rightarrow b_n = \frac{2}{n}, n = 1, 2, 3, 4, \dots$$

$$b_n = \frac{2}{n}, n = 1, 2, 3, 4, \dots$$

-----→(6b)

Thus substituting the values of a_0, a_n & b_n from eqns. (4), (5) & (6b) respectively in the Fourier series expansion of $f(x)$ over the interval $(c, c + 2l)$ we get

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right), c = 0 \text{ \& } l = \pi$$

$$\Rightarrow f(x) = 0 + 0 + \sum_{n=1}^{\infty} \left\{ \left(\frac{2}{n} \right) * \sin\left(\frac{n\pi x}{\pi}\right) \right\}$$

$$\therefore (\pi - x) = 2 * \sum_{n=1}^{\infty} \frac{\sin(nx)}{n} \quad \text{-----→(7)}$$

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$$\therefore (\pi - x) = 2 * \sum_{n=1}^{\infty} \frac{\sin(nx)}{n}$$

----->(7)

Thus eqn.(7) gives the Fourier series expansion of $f(x) = (\pi - x)$ over the interval $(0, 2\pi)$

Deduction:

Put $x = \frac{\pi}{2} \in (0, 2\pi)$ on both the sides of eqn.(7) we get

$$\begin{aligned} \left(\pi - \frac{\pi}{2}\right) &= 2 * \sum_{n=1}^{\infty} \frac{\sin\left(\frac{n\pi}{2}\right)}{n} \\ \Rightarrow \frac{\pi}{2} &= 2 * \left\{ \frac{\sin\left(\frac{\pi}{2}\right)}{1} + \frac{\sin\left(\frac{2\pi}{2}\right)}{2} + \frac{\sin\left(\frac{3\pi}{2}\right)}{3} + \frac{\sin\left(\frac{4\pi}{2}\right)}{4} + \frac{\sin\left(\frac{5\pi}{2}\right)}{5} + \frac{\sin\left(\frac{6\pi}{2}\right)}{6} + \frac{\sin\left(\frac{7\pi}{2}\right)}{7} + \dots \right\} \end{aligned}$$

$$\Rightarrow \frac{\pi}{2} = 2 * \left\{ 1 + 0 + \frac{(-1)}{3} + 0 + \frac{1}{5} + 0 + \frac{(-1)}{7} + - \dots \right\}$$

$$\Rightarrow \frac{\pi}{2} = 2 * \left\{ 1 + -\frac{1}{3} + \frac{1}{5} - \frac{1}{7} + - \dots \right\} \quad \Rightarrow \frac{\pi}{4} = 1 + -\frac{1}{3} + \frac{1}{5} - \frac{1}{7} + - \dots$$

$$\Rightarrow 1 + -\frac{1}{3} + \frac{1}{5} - \frac{1}{7} + - \dots = \frac{\pi}{4}$$

$$\therefore \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n-1} = \frac{\pi}{4} \quad \text{(or)} \quad 1 + -\frac{1}{3} + \frac{1}{5} - \frac{1}{7} + - \dots = \frac{\pi}{4}$$

3) Obtain the Fourier series expansion of $f(x) = x \cos(x)$ in $-\pi < x < \pi$

Soltn: By data we have $f(x) = x \cos(x)$ ----->(1)

& $x \in (-\pi, \pi) = (c, c + 2l)$. On comparing we get

$$c = -\pi \quad \& \quad l = \pi \quad \text{----->(2)}$$

Nature of the function:

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Wkt, $x \rightarrow \text{odd fn}$ & $\cos(x) \rightarrow \text{even fn}$.

Wkt, the product of an odd fn with an even fn is always an ODD fn.

$\therefore f(x) = x \cos(x)$ is an **ODD** function in $-\pi < x < \pi$

$$a_0 = 0$$

-----→(3)

$$a_n = 0$$

-----→(4)

To find b_n consider

$$b_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx, \quad c = -\pi \quad \& \quad l = \pi$$

$$\Rightarrow b_n = \frac{1}{\pi} \int_{x=-\pi}^{\pi} f(x) \sin\left(\frac{n\pi x}{\pi}\right) dx \Rightarrow b_n = \frac{1}{\pi} \int_{x=-\pi}^{\pi} f(x) \sin(nx) dx$$

Where, $f(x) = x \cos(x)$ is an ODD function

$$b_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin(nx) dx$$

$$b_n = \frac{2}{\pi} \int_{x=0}^{\pi} x \cos(x) \sin(nx) dx \quad \text{-----→(5a)}$$

$$\text{Wkt } \cos A \sin B = \frac{1}{2} \{ \sin(A+B) - \sin(A-B) \}, \quad A = x \quad \& \quad B = nx$$

$$A+B = x + nx = x(1+n) \quad (\text{or}) \quad (n+1)x$$

$$A-B = x - nx = x(1-n) \quad (\text{or}) \quad -(n-1)x$$

$$\cos x \sin(nx) = \frac{1}{2} \{ \sin(n+1)x - \sin(-(n-1)x) \}$$

(Note: $\cos x \sin(nx) = \sin(nx) \cos(x) \because$ angles for cosine & sine are same on both sides but $\cos x \sin(nx) \neq \sin x \cos(nx)$)

$$\cos A * \sin B = \sin B * \cos A \quad \text{but} \quad \cos A * \sin B \neq \sin A * \cos B \quad]$$

$$\cos x \sin(nx) = \frac{1}{2} \{ \sin(n+1)x + \sin((n-1)x) \} \because \sin(-\theta) = -\sin(\theta) \quad \text{-----→(5b)}$$

Substitute eqn. (5b) in RHS of eqn.(5a) we get



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$$\begin{aligned}
 b_n &= \frac{2}{\pi} \int_{x=0}^{\pi} x \frac{1}{2} \{ \sin(n+1)x + \sin((n-1)x) \} dx \\
 \Rightarrow b_n &= \frac{2}{2\pi} \int_{x=0}^{\pi} x \{ \sin(n+1)x + \sin((n-1)x) \} dx \\
 \Rightarrow b_n &= \frac{1}{\pi} \int_{x=0}^{\pi} x \{ \sin(n+1)x + \sin((n-1)x) \} dx \\
 \Rightarrow b_n &= \frac{1}{\pi} \left\{ \int_{x=0}^{\pi} x * \sin(n+1)x dx + \int_{x=0}^{\pi} x * \sin(n-1)x dx \right\} \quad \text{-----} \rightarrow (5c)
 \end{aligned}$$

By applying Bernoulli's rule of integration by parts,

$$\begin{aligned}
 &\int_{x=0}^{\pi} x * \sin(n+1)x dx \\
 &= \left[\left\{ x \cdot \frac{-\cos(n+1)x}{(n+1)} \right\} - \left\{ (1) \cdot \frac{-\sin(n+1)x}{(n+1)^2} \right\} + 0 \right]_0^{\pi} \\
 &= \left[\left\{ x \cdot \frac{-\cos(n+1)x}{(n+1)} \right\} \right]_0^{\pi} + 0 \quad \because \sin(n+1)\pi = \sin(0) = 0 \\
 &= \frac{-1}{(n+1)} [\{x \cdot \cos(n+1)x\}]_0^{\pi} \\
 &= \frac{-1}{(n+1)} [\{\pi \cdot \cos((n+1)\pi)\} - \{0\}] \\
 &= \frac{-1}{(n+1)} \{\pi * (-1)^{n+1}\} \quad \because \cos(n\pi) = (-1)^n \\
 &= \frac{-1}{(n+1)} \{\pi * (-1) (-1)^n\} \quad (or) \quad \pi \frac{(-1)^n}{(n+1)} \\
 \therefore \int_{x=0}^{\pi} x * \sin(n+1)x dx &= -\pi \frac{(-1)^{n+1}}{(n+1)} \quad \text{-----} \rightarrow (5d) \\
 \sim ly \quad \therefore \int_{x=0}^{\pi} x * \sin(n-1)x dx &= -\pi \frac{(-1)^{n-1}}{(n-1)} \quad \text{-----} \rightarrow (5e)
 \end{aligned}$$



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Substituting eqns. (5d) & (5e) in the RHS of eqn.(5c) we get

$$b_n = \frac{1}{\pi} \left\{ \left[-\pi \frac{(-1)^{n+1}}{(n+1)} \right] + \left[-\pi \frac{(-1)^{n-1}}{(n-1)} \right] \right\}$$

$$b_n = \frac{-\pi}{\pi} \left\{ \left[\frac{(-1)^{n+1}}{(n+1)} \right] + \left[\frac{(-1)^{n-1}}{(n-1)} \right] \right\}$$

$$b_n = -1 \left\{ \left[\frac{(-1)(-1)^n}{(n+1)} \right] + \left[\frac{(-1)(-1)^n}{(n-1)} \right] \right\}$$

$$b_n = (-1)(-1)(-1)^n \left\{ \frac{1}{(n+1)} + \frac{1}{(n-1)} \right\}$$

$$b_n = (-1)^n \left\{ \frac{1}{(n+1)} + \frac{1}{(n-1)} \right\}$$

$$b_n = (-1)^n \left\{ \frac{2n}{n^2 - 1} \right\} \quad (or) \quad b_n = 2 \left\{ \frac{(-1)^n * n}{n^2 - 1} \right\}$$

$$\therefore b_n = 2 \left\{ \frac{(-1)^n * n}{n^2 - 1} \right\}, \quad n \neq 1 \quad \text{-----} \rightarrow (6)$$

To find b_n at $n = 1$ i.e b_1 put $n = 1$ in eqn.(5a)

$$b_1 = \frac{2}{\pi} \int_{x=0}^{\pi} x \cos(x) \sin(x) dx$$

$$\text{Wkt } \sin(2x) = 2\sin(x)\cos(x) = 2\cos(x)\sin(x) \therefore \cos(x)\sin(x) = \frac{\sin(2x)}{2}$$

$$\Rightarrow b_1 = \frac{2}{\pi} \int_{x=0}^{\pi} x \frac{\sin(2x)}{2} dx \quad \Rightarrow b_1 = \frac{2}{2\pi} \int_{x=0}^{\pi} x * \sin(2x) dx$$

By applying Bernoulli's rule of integration by parts, $u = u_0 = x$ & $v = \sin(2x)$

$$\Rightarrow b_1 = \frac{1}{\pi} \left[\left\{ x * \frac{-\cos(2x)}{2} \right\} - \left\{ 1 * \frac{-\sin(2x)}{4} \right\} + 0 \right]_0^{\pi}$$

$$\Rightarrow b_1 = \frac{1}{\pi} \left[\frac{-1}{2} \{ \pi * \cos(2\pi) - 0 \} + \frac{1}{4} \{ 0 - 0 \} \right] \because \sin(2\pi) = \sin(0) = 0$$

$$\Rightarrow b_1 = \frac{1}{\pi} \left[\frac{-1}{2} \{ \pi * 1 \} \right] \because \cos(2\pi) = 1$$

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$$\Rightarrow b_1 = \frac{1}{\pi} * \left(\frac{-\pi}{2} \right) \therefore b_1 = -\left(\frac{1}{2} \right)$$

$b_1 = -\left(\frac{1}{2} \right)$

----->(7)

Thus substituting the above obtained values of a_0 , a_n , b_n & b_1 from eqns.(3), (4), (6) & (7) respectively in the Fourier series expansion of

$f(x) = x * \cos x$ in $(-\pi, \pi)$, $c = -\pi$ & $l = \pi$ we get

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right), \quad c = -\pi \text{ & } l = \pi$$

$$x * \cos x = 0 + \sum_{n=1}^{\infty} 0 + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{\pi}\right)$$

$$x * \cos x = \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$\Rightarrow x * \cos x = b_1 \sin(x) + \sum_{n=2}^{\infty} b_n \sin(nx)$$

$\therefore x * \cos x = \left(\frac{-1}{2} \right) \sin(x) + 2 \sum_{n=2}^{\infty} \left\{ \frac{(-1)^n * n}{n^2 - 1} \right\} \sin(nx)$

----->(8)

Thus eqn.(8) gives the Fourier series expansion of $f(x) = x * \cos(x)$ in $-\pi < x < \pi$

4) Find the Fourier series expansion of $f(x) = x^2$ in $-\pi < x < \pi$.

Soltn: By data, $f(x) = x^2$ ----->(1) $x \in (-\pi, \pi) = (c, c + 2l)$

$$c = -\pi \text{ & } l = \pi \text{ ----->(2)}$$

Nature of the function in $(-\pi, \pi)$:

$$f(-x) = (-x)^2 \Rightarrow f(-x) = x^2 \therefore f(-x) = f(x)$$

$\therefore f(x) = x^2$ is an **even** function in $(-\pi, \pi)$

$$\therefore b_n = 0 \text{ ----->(3)}$$

$$\text{Wkt } a_0 = \frac{2}{l} \int_{x=0}^l f(x) dx$$



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$$a_0 = \frac{2}{\pi} \int_{x=0}^{\pi} x^2 dx \quad \Rightarrow \quad a_0 = \frac{2}{\pi} * \left[\frac{x^3}{3} \right]_0^{\pi}$$

$$a_0 = \frac{2\pi^2}{3} \quad (or) \quad \frac{a_0}{2} = \frac{\pi^2}{3} \quad \text{-----} \rightarrow (4)$$

$$\text{Wkt, } a_n = \frac{2}{l} \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad l = \pi$$

$$a_n = \frac{2}{\pi} \int_{x=0}^{\pi} x^2 \cos(nx) dx$$

$$a_n = \frac{2}{\pi} \left[\left\{ x^2 * \left(\frac{\sin(nx)}{n} \right) \right\} - \left\{ 2x * \left(\frac{-\cos(nx)}{n^2} \right) \right\} + \left\{ 2 * \left(\frac{-\sin(nx)}{n^3} \right) \right\} - 0 \right]_0^{\pi}$$

$$a_n = \frac{2}{\pi} \left[- \left\{ 2x * \left(\frac{-\cos(nx)}{n^2} \right) \right\} \right]_0^{\pi} \quad \because \sin(n\pi) = 0 = \sin(0)$$

$$a_n = \frac{2}{\pi} * \frac{2}{n^2} [x * \cos(nx)]_0^{\pi} \Rightarrow a_n = \frac{4}{\pi n^2} [\pi(-1)^n] \quad \because \cos(n\pi) = (-1)^n \text{ \& } \cos(0) = 1$$

$$a_n = 4 * \frac{(-1)^n}{n^2} \quad \text{-----} \rightarrow (5)$$

Thus substituting the above obtained values of a_0 , a_n , b_n from eqns.(4), (5), (3) respectively in the Fourier series expansion of

$f(x) = x^2$ in $(-\pi, \pi)$, $c = -\pi$ & $l = \pi$ we get

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right), \quad c = -\pi \text{ \& } l = \pi$$

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx)$$

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5) Find the Fourier series expansion of $f(x) = x$ in $-\pi < x < \pi$.

Soltn: By data, $f(x) = x \rightarrow (1) \quad x \in (-\pi, \pi) = (c, c + 2l)$

$$c = -\pi \text{ \& } l = \pi \rightarrow (2)$$

Nature of the function in $(-\pi, \pi)$:

$$f(-x) = -x \quad \therefore f(-x) = -f(x)$$

$\therefore f(x) = x$ is an **odd** function in $(-\pi, \pi)$

$$\therefore a_0 = 0 \rightarrow (3) \text{ \& }$$

$$\therefore a_n = 0 \rightarrow (4)$$

$$\text{Wkt } b_n = \frac{2}{l} \int_{x=0}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx, \quad l = \pi$$

$$b_n = \frac{2}{\pi} \int_{x=0}^{\pi} x \sin(nx) dx$$

$$b_n = \frac{2}{\pi} \left[\left\{ x * \left(\frac{-\cos(nx)}{n} \right) \right\} - \left\{ 1 * \left(\frac{-\sin(nx)}{n^2} \right) \right\} + 0 \right]_0^{\pi}$$

$$b_n = \frac{2}{\pi} * \left(\frac{-1}{n} \right) [x * \cos(nx)]_0^{\pi} \because \sin(n\pi) = 0 = \sin(0)$$

$$b_n = \left(\frac{-2}{n\pi} \right) [\pi(-1)^n] \because \cos(n\pi) = (-1)^n$$

$$\therefore b_n = 2 * \left\{ \frac{(-1)^{n+1}}{n} \right\} \rightarrow (5)$$

Thus substituting the above obtained values of a_0 , a_n , b_n from eqns.(3), (4), (5) respectively in the Fourier series expansion of

$f(x) = x$ in $(-\pi, \pi)$, $c = -\pi$ & $l = \pi$ we get

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right), \quad c = -\pi \text{ \& } l = \pi$$

$$x = 2 * \sum_{n=1}^{\infty} \left\{ \frac{(-1)^{n+1}}{n} \right\} \sin(nx)$$

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Interval $(0, 2l)$ & $(-l, l)$

6) Find the Fourier series expansion of $f(x) = e^x$ in $-l < x < l$.

Solt: By data we have, $f(x) = e^x$ -----→(1)

$$x \in (-l, l) = (c, c + 2l) \quad \therefore c = -l \quad \text{-----→(2)}$$

Nature of the function in $(-l, l)$:

$$f(-x) = e^{-x} \Rightarrow f(-x) \neq f(x) \text{ \& } f(-x) \neq -f(x)$$

$f(x) = e^x$ is Neither Even Nor Odd in the $(-l, l)$

$$a_0 = \frac{1}{l} \int_{x=c}^{c+2l} f(x) dx, c = -l$$

$$a_0 = \frac{1}{l} \int_{x=-l}^l e^x dx \quad \therefore a_0 = \frac{2\sinh(l)}{l} \quad (\text{or}) \quad \frac{a_0}{2} = \frac{\sinh(l)}{l} \quad \text{-----→(3)}$$

$$a_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx, c = -l$$

$$a_n = \frac{1}{l} \int_{x=-l}^l e^x \cos\left(\frac{n\pi x}{l}\right) dx,$$

wkt

$$\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} \{a \cos bx + b \sin bx\}, a = 1 \text{ \& } b = \frac{n\pi}{l} \text{ are constants}$$

$$a_n = \frac{1}{l} * \left\{ \frac{l^2}{l^2 + n^2 \pi^2} \right\} * \{(-1)^n * 2\sinh(l)\}$$

$$\therefore a_n = l * 2\sinh(l) * \left\{ \frac{(-1)^n}{l^2 + n^2 \pi^2} \right\} \quad \text{-----→(4)}$$

$$b_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx, c = -l$$

$$b_n = \frac{1}{l} \int_{x=-l}^l e^x \sin\left(\frac{n\pi x}{l}\right) dx$$

$$\text{Wkt } \int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} \{a \sin bx - b \cos bx\}, a = 1 \text{ \& } b = \frac{n\pi}{l} \text{ are constants}$$

$$b_n = \frac{1}{l^2} * \left\{ \frac{-l^2}{l^2 + n^2 \pi^2} \right\} * \pi \{(-1)^n * n * 2\sinh(l)\}$$

$$b_n = \pi * 2\sinh(l) * \left\{ \frac{(-1)^{n+1} * n}{l^2 + n^2 \pi^2} \right\} \quad \text{-----→(5)}$$

$$e^x = \frac{\sinh(l)}{l} + 2l * \sinh(l) \sum_{n=1}^{\infty} \frac{(-1)^n}{l^2 + n^2 \pi^2} * \cos\left(\frac{n\pi x}{l}\right) +$$

$$2\pi \sinh(l) \sum_{n=1}^{\infty} \frac{(-1)^{n+1} * n}{l^2 + n^2 \pi^2} * \sin\left(\frac{n\pi x}{l}\right)$$

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7) Obtain the Fourier series expansion of the function $f(x) = e^x, 0 < x < 2l$

Soltn: By data we have, $f(x) = e^x \rightarrow (1)$

$$x \in (0, 2l) = (c, c + 2l) \therefore c = 0 \rightarrow (2)$$

Nature of the function in $(0, 2l)$:

$$f(2l - x) = e^{2l-x} \Rightarrow f(2l - x) \neq f(x) \text{ \& } f(2l - x) \neq -f(x)$$

$f(x) = e^x$ is Neither Even Nor Odd in the $(0, 2l)$

$$a_0 = \frac{1}{l} \int_{x=c}^{c+2l} f(x) dx, c = 0$$

$$a_0 = \frac{1}{l} \int_{x=0}^{2l} e^x dx \therefore a_0 = \frac{e^{2l}-1}{l} \text{ (or) } \frac{a_0}{2} = \frac{e^{2l}-1}{2l} \rightarrow (3)$$

$$a_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx, c = 0$$

$$a_n = \frac{1}{l} \int_{x=0}^{2l} e^x \cos\left(\frac{n\pi x}{l}\right) dx$$

wkt

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2+b^2} \{a \cos bx + b \sin bx\}, a = 1 \text{ \& } b = \frac{n\pi}{l} \text{ are constants}$$

$$a_n = \left\{ \frac{l}{l^2+n^2\pi^2} \right\} * (e^{2l} - 1)$$

$$\therefore a_n = l * (e^{2l} - 1) * \left\{ \frac{1}{l^2+n^2\pi^2} \right\} \rightarrow (4)$$

$$b_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx, c = 0$$

$$b_n = \frac{1}{l} \int_{x=0}^{2l} e^x \sin\left(\frac{n\pi x}{l}\right) dx$$

$$\text{Wkt } \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2+b^2} \{a \sin bx - b \cos bx\}, a = 1 \text{ \& } b = \frac{n\pi}{l} \text{ are constants}$$

$$b_n = \frac{1}{l^2} * \left\{ \frac{-l^2}{l^2+n^2\pi^2} \right\} * \frac{\pi}{l} \{n * (e^{2l} - 1)\}$$

$$b_n = \pi * (1 - e^{2l}) \left\{ \frac{n}{l^2+n^2\pi^2} \right\} \rightarrow (5)$$

$$e^x = \frac{e^{2l}-1}{2l} + l(e^{2l} - 1) \sum_{n=1}^{\infty} \frac{\cos\left(\frac{n\pi x}{l}\right)}{l^2+n^2\pi^2} + \pi(1 - e^{2l}) \sum_{n=1}^{\infty} n * \frac{\sin\left(\frac{n\pi x}{l}\right)}{l^2+n^2\pi^2}$$

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Note:

- 1) If $f(x) = \begin{cases} \phi(x), & -\pi < x < 0 \\ \psi(x), & 0 < x < \pi \end{cases}$ then we say that $f(x)$ in the interval $(-\pi, \pi)$ is

a) **Even** function: when $\phi(-x) = \psi(x)$

b) **Odd** function : when $\phi(-x) = -\psi(x)$

The above definition can be extended to the interval $(-l, l)$

Examples:

- 1) $f(x) = \begin{cases} -x, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$ is an **even** function because

Here $\phi(x) = -x$ & $\psi(x) = x \quad \therefore \phi(-x) = -(-x) \Rightarrow \phi(-x) = x$
 $\therefore \phi(-x) = \psi(x) \quad \therefore f(x)$ is an even function in interval $(-\pi, \pi)$

- 2) $f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & -\pi < x < 0 \\ 1 - \frac{2x}{\pi}, & 0 < x < \pi \end{cases}$ is an **even** function because

Here $\phi(x) = 1 + \frac{2x}{\pi}$ & $\psi(x) = 1 - \frac{2x}{\pi}$

$\therefore \phi(-x) = 1 - \frac{2x}{\pi} \Rightarrow \phi(-x) = \psi(x)$

Thus $f(x)$ is an **even** function in the interval $(-\pi, \pi)$

- 3) $f(x) = \begin{cases} x - \frac{\pi}{2}, & -\pi < x < 0 \\ x + \frac{\pi}{2}, & 0 < x < \pi \end{cases}$ is an **odd** function because

Here $\phi(x) = x - \frac{\pi}{2}$ & $\psi(x) = x + \frac{\pi}{2}$

$\therefore \phi(-x) = -x - \frac{\pi}{2} \Rightarrow \phi(-x) = -\left(x + \frac{\pi}{2}\right)$



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$$\therefore \phi(-x) = -\psi(x)$$

Thus $f(x)$ is an **odd** function in the interval $(-\pi, \pi)$

4) $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ +k, & 0 < x < \pi \end{cases}$ is an **odd** function because

$$\text{Here } \phi(x) = -k \text{ \& } \psi(x) = +k$$

$$\therefore \phi(-x) = -k \Rightarrow \phi(-x) = -\psi(x)$$

$$\therefore \phi(-x) = -\psi(x)$$

Thus $f(x)$ is an **odd** function in the interval $(-\pi, \pi)$

Note: $f(x) = |-x|$ is an even function because $|-x| = |x|$

$$\text{Wkt } |-x| = \begin{cases} -x, & x < 0 \\ +x, & x > 0 \end{cases}$$

Similarly,

If $f(x) = \begin{cases} \phi(x), & 0 < x < \pi \\ \psi(x), & \pi < x < 2\pi \end{cases}$ then we say that $f(x)$ in the interval $(0, 2\pi)$ is

a) **Even** function: when $\phi(2\pi - x) = \psi(x)$

b) **Odd** function: when $\phi(2\pi - x) = -\psi(x)$

The above definition can be extended to the interval $(0, 2l)$

a) **Even** function: when $\phi(2l - x) = \psi(x)$

b) **Odd** function: when $\phi(2l - x) = -\psi(x)$

8) Obtain the Fourier series for the function $f(x) = 2x - x^2$ in $0 \leq x \leq 2$

Solt: By data $f(x) = 2x - x^2$ -----→(1)

$x \in [0, 2]$ but $(0, 2) \in [0, 2] \therefore x \in (0, 2)$ -----→(2)

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Comparing $(0, 2)$ with $(c, c + 2l)$ we get $c = 0$ & $l = 1$

$$\boxed{c = 0 \text{ \& } l = 1} \quad \text{-----} \rightarrow (3)$$

Nature of the function: in the interval $(0, 2l)$, $l = 1$ consider

$$f(2l - x) = 2(2l - x) - (2l - x)^2, \quad l = 1$$

$$f(2 - x) = 2(2 - x) - (2 - x)^2 \Rightarrow f(2 - x) = (2 - x)\{2 - (2 - x)\}$$

$$\Rightarrow f(2 - x) = (2 - x)\{2 - 2 + x\} \Rightarrow f(2 - x) = (2 - x)(x)$$

$$\Rightarrow f(2 - x) = 2x - x^2 \Rightarrow f(2 - x) = f(x)$$

$\therefore f(x) = 2x - x^2$ is an **even** function in the interval $(0, 2)$

$$\boxed{\therefore b_n = 0} \quad \text{-----} \rightarrow (4)$$

$$\text{Wkt } a_0 = \frac{1}{l} \int_c^{c+2l} f(x) dx, \quad c = 0 \text{ \& } l = 1$$

$$\therefore a_0 = \frac{1}{1} \int_0^2 f(x) dx, \quad f(x) \rightarrow \text{even fn.}$$

$$\therefore a_0 = 2 \int_0^1 f(x) dx \Rightarrow \therefore a_0 = 2 \int_0^1 \{2x - x^2\} dx$$

$$\therefore a_0 = 2 \left[2 \left(\frac{x^2}{2} \right) - \left(\frac{x^3}{3} \right) \right]_{x=0}^1$$

$$\therefore a_0 = 2 \left[\left\{ 2 \left(\frac{1}{2} \right) - \left(\frac{1}{3} \right) \right\} - 0 \right] \Rightarrow \therefore a_0 = 2 \left[1 - \frac{1}{3} \right]$$

$$\therefore a_0 = 2 * \frac{2}{3} \quad \therefore a_0 = \frac{4}{3} \quad (\text{or}) \quad \frac{a_0}{2} = \frac{2}{3}$$

$$\boxed{\therefore a_0 = \frac{4}{3} \quad (\text{or}) \quad \frac{a_0}{2} = \frac{2}{3}} \quad \text{-----} \rightarrow (5)$$

$$\text{Wkt } a_n = \frac{2}{l} \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad l = 1$$

$$a_n = \frac{2}{1} \int_{x=0}^1 \{2x - x^2\} \cos\left(\frac{n\pi x}{1}\right) dx \Rightarrow a_n = 2 \int_{x=0}^1 \{2x - x^2\} \cos(n\pi x) dx$$

$$\text{-----} \rightarrow (6a)$$

By applying Bernoulli's rule of integration by parts,

$$\int uv = u_0 v_1 - u_1 v_2 + u_2 v_3 - u_3 v_4 + \dots$$

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$u = u_0 = 2x - x^2$	$v = \cos(n\pi x)$
$u_1 = u' = 2 - 2x$	$\therefore v_1 = \int \cos(n\pi x) dx \therefore v_1 = \frac{\sin(n\pi x)}{n\pi}$
$u_2 = u'' = -2$	$\therefore v_2 = \int \frac{\sin(n\pi x)}{n\pi} dx \therefore v_2 = \frac{-\cos(n\pi x)}{(n\pi)^2}$
$u_3 = u''' = 0$	$\therefore v_3 = \int \frac{-\cos(n\pi x)}{(n\pi)^2} dx \therefore v_3 = \frac{-\sin(n\pi x)}{(n\pi)^3}$

Using above obtained values in the RHS of eqn. (6a) we get

$$a_n = 2 \left[\left\{ (2x - x^2) * \frac{\sin(n\pi x)}{n\pi} \right\} - \left\{ (2 - 2x) * \left(\frac{-\cos(n\pi x)}{(n\pi)^2} \right) \right\} + \left\{ (-2) * \left(\frac{-\sin(n\pi x)}{(n\pi)^3} \right) \right\} \right]_0^1$$

$$a_n = 2 \left[\{0 - 0\} + \left(\frac{1}{(n\pi)^2} \right) \{ (2 - 2x) * \cos(n\pi x) \}_0^1 + \{0 - 0\} \right] \therefore \sin(0) = \sin(n\pi) = 0$$

$$a_n = 2 \left[\left(\frac{2}{(n\pi)^2} \right) \{ (1 - x) * \cos(n\pi x) \} \right]_0^1$$

$$a_n = \left(\frac{4}{(n\pi)^2} \right) \{0 - 1\} \therefore \cos(0) = 1 \therefore a_n = \frac{-4}{n^2 \pi^2}$$

$\therefore a_n = \frac{-4}{n^2 \pi^2}$

.....>(6b)

Substituting the values of a_0 , a_n & b_n from eqns. (5), (6b) & (4) respectively in the Fourier series expansion $f(x)$ over interval $(c, c + 2l) = (0, 2), c = 0$ & $l = 1$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right), c = 0 \text{ \& \;} l = 1$$

$$\Rightarrow f(x) = \frac{2}{3} + \sum_{n=1}^{\infty} \left(\frac{-4}{n^2 \pi^2} \right) \cos(n\pi x) + 0$$

$\therefore 2x - x^2 = \frac{2}{3} - \left(\frac{4}{\pi^2} \right) * \sum_{n=1}^{\infty} \left[\frac{\cos(n\pi x)}{n^2} \right]$

----->(7)

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Gives the Fourier series expansion of $f(x) = 2x - x^2$ in $0 \leq x \leq 2$

- 9) Draw the graph of the function $f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1 \\ \pi(2-x), & 1 \leq x \leq 2 \end{cases}$ and also

Show that the Fourier expansion of the function $f(x)$ is

$$\frac{\pi}{2} - \frac{4}{\pi} \left\{ \frac{\cos \pi x}{1^2} - \frac{\cos 3\pi x}{3^2} + \frac{\cos 5\pi x}{5^2} + \dots \right\}$$

Hence show that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

Solt: By data, $f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1 \\ \pi(2-x), & 1 \leq x \leq 2 \end{cases} \dots \dots \dots \rightarrow (1)$

Here interval is $(c, c+2l) = (0, 2) \therefore c = 0 \text{ \& } l = 1$

$c = 0 \text{ \& } l = 1$

$\dots \dots \dots \rightarrow (2)$

Comparing eqn.(1) to the standard form $f(x) = \begin{cases} \phi(x), & 0 < x < l \\ \psi(x), & l < x < 2l \end{cases}$

we get $\phi(x) = \pi x$ & $\psi(x) = \pi(2-x)$

$$\phi(2l-x) = \pi(2l-x), \quad l=1 \Rightarrow \phi(2-x) \neq \psi(x) \text{ \& } \phi(2\pi-x) \neq -\psi(x)$$

$$\phi(2-x) = \pi(2-x) \Rightarrow \phi(2-x) = \psi(x)$$

Therefore the given function $f(x)$ is an **even** function in the interval $(0, 2)$

$\therefore b_0 = 0$

$\dots \dots \dots \rightarrow (3)$

Euler's co-efficients:

1) $a_0 = \frac{2}{l} \int_{x=0}^l f(x) dx, \quad f(x) \text{ is an even function } c = 0 \text{ \& } l = 1$

$$\Rightarrow a_0 = \frac{2}{1} \int_{x=0}^1 f(x) dx$$

$$\Rightarrow a_0 = 2 \int_{x=0}^1 \pi x dx, \quad \text{using eqn.(1)}$$

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$$\Rightarrow a_0 = 2\pi \left\{ \frac{x^2}{2} \right\}_0^1 \Rightarrow a_0 = \frac{\pi}{1} \{1 - 0\}$$

$\therefore a_0 = \pi \quad (\text{or}) \quad \frac{a_0}{2} = \frac{\pi}{2}$

-----→(4)

2) $a_n = \frac{1}{l} \int_{x=c}^{c+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad c = 0 \text{ \& } l = 1$

$$\Rightarrow a_n = \frac{1}{l} \int_{x=0}^{2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$\Rightarrow a_n = \frac{2}{l} \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \because f(x) \text{ is an even fn.}$$

$$\Rightarrow a_n = \frac{2}{1} \int_{x=0}^1 (\pi x) \cos\left(\frac{n\pi x}{1}\right) dx$$

$$\Rightarrow a_n = 2 \int_{x=0}^1 (\pi x) \cos(n\pi x) dx$$

$$\Rightarrow a_n = 2\pi \int_{x=0}^1 (x) \cos(n\pi x) dx \quad \text{-----→(5a)}$$

By Bernoulli's rule of integration of parts

$$\Rightarrow a_n = 2\pi \left\{ \left[x \left(\frac{\sin(n\pi x)}{n\pi} \right) \right] - \left[(1) \left(\frac{-\cos(n\pi x)}{(n\pi)^2} \right) \right] + 0 \right\}_0^1$$

$$\Rightarrow a_n = 2\pi \left\{ \left[x \left(\frac{\sin(n\pi x)}{n\pi} \right) \right] + \left[(1) \left(\frac{\cos(n\pi x)}{(n\pi)^2} \right) \right] \right\}_0^1$$

$$\Rightarrow a_n = 2\pi \left\{ [0 - 0] + \frac{1}{(n\pi)^2} [\cos(n\pi) - 1] \right\} \because \sin(0) = \sin(n\pi) = 0$$

$$\Rightarrow a_n = \frac{2\pi}{n^2 \pi^2} \{\cos(n\pi) - 1\} \text{ \& wkt } \cos(n\pi) = (-1)^n$$

$\therefore a_n = \frac{2}{n^2 \pi} \{(-1)^n - 1\}$

-----→(5b)

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Substituting the above obtained values of a_0 , a_n & b_n from equations (4), (5b) and (3) respectively in the Fourier series expansion of $f(x)$ in $(c, c + 2l)$, $c = 0$ & $l = 1$ i.e $f(x)$ in $(0, 2)$ we get

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right), \quad c = 0 \text{ \& \;} l = 1$$

$$\Rightarrow f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left\{ \frac{2}{n^2\pi} \{(-1)^n - 1\} \right\} \cos(n\pi x) + 0$$

$$\Rightarrow f(x) = \frac{\pi}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \left\{ \frac{(-1)^n - 1}{n^2} \right\} \cos(n\pi x)$$

$$\Rightarrow f(x) = \frac{\pi}{2} + \frac{2}{\pi} \left\{ \frac{-2}{1^2} * \cos(\pi x) + 0 + \frac{-2}{3^2} * \cos(3\pi x) + 0 + \frac{-2}{5^2} * \cos(5\pi x) + \dots \right\}$$

$$\Rightarrow f(x) = \frac{\pi}{2} + \frac{2}{\pi} \left\{ \frac{-2}{1^2} * \cos(\pi x) + \frac{-2}{3^2} * \cos(3\pi x) + \frac{-2}{5^2} * \cos(5\pi x) + \dots \right\}$$

$$\Rightarrow f(x) = \frac{\pi}{2} + \frac{-4}{\pi} \left\{ \frac{\cos(\pi x)}{1^2} + \frac{\cos(3\pi x)}{3^2} + \frac{\cos(5\pi x)}{5^2} + \dots \right\}$$

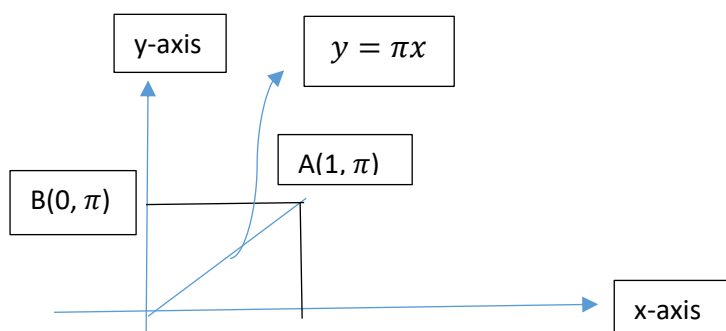
$$\therefore f(x) = \frac{\pi}{2} - \frac{4}{\pi} \left\{ \frac{\cos(\pi x)}{1^2} + \frac{\cos(3\pi x)}{3^2} + \frac{\cos(5\pi x)}{5^2} + \dots \right\}$$

-----→(6)

Equation (6) gives the Fourier series expansion of $f(x)$ in $(0, 2)$

Graph:

i) $y = \pi x$ ($y = mx$, $m = \tan\theta = \pi$ & $opp = \pi$, $adj = 1$)

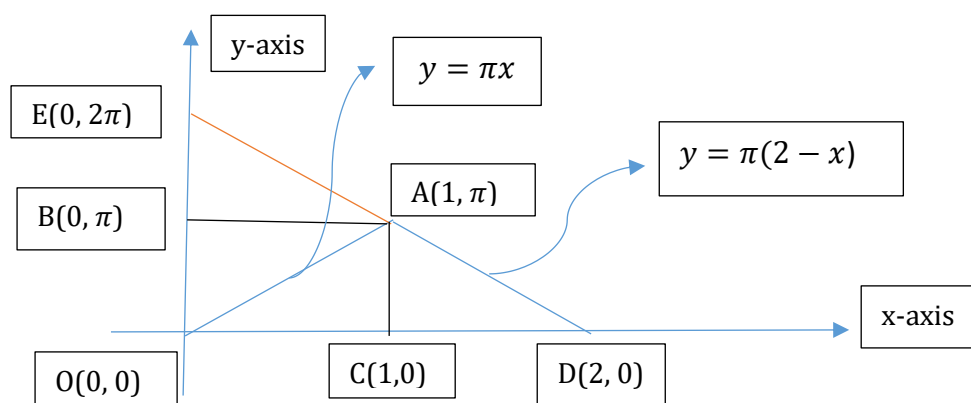


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O(0, 0)

C(1,0)

ii) $y = \pi(2 - x) \Rightarrow y = 2\pi - \pi x \Rightarrow \pi x + y = 2\pi \therefore \frac{x}{2} + \frac{y}{2\pi} = 1$



Note: In the solution for the problem given by eqn.(6)

Let $x = 0 \in [0, 1] \therefore f(x) = 0$, using eqn. (1)

Put $f(x) = 0$ in LHS of eqn. (6) & $x = 0$ in RHS of eqn. (6) we get

$$0 = \frac{\pi}{2} - \frac{4}{\pi} \left\{ \frac{\cos(0)}{1^2} + \frac{\cos(0)}{3^2} + \frac{\cos(0)}{5^2} + \dots \right\}$$

$$\Rightarrow -\frac{\pi}{2} = -\frac{4}{\pi} \left\{ \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right\} \therefore \cos(0) = 1$$

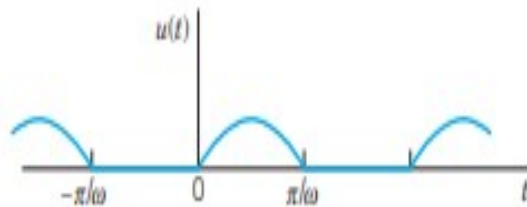
$$\therefore \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \quad \text{-----} \rightarrow (7)$$



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- 10) A sinusoidal voltage (ωt) , t is time is passed through a half wave rectifier that clips the negative portion of the wave as shown in the figure below. Find the Fourier series of the resulting periodic function

$$u(t) = \begin{cases} 0, & -L < t < 0 \\ E \sin(\omega t), & 0 < t < L \end{cases} \quad \text{where period } p = \frac{2\pi}{\omega}$$



Soltn: By data, $u(t) = \begin{cases} 0, & -L < t < 0 \\ E \sin(\omega t), & 0 < t < L \end{cases}$

For simplicity let $u(t)$ be replaced by $f(t)$

$$\therefore f(t) = \begin{cases} 0, & -L < t < 0 \\ E \sin(\omega t), & 0 < t < L \end{cases} \quad \text{-----} \rightarrow (1)$$

Here $p = 2L = \frac{2\pi}{\omega} \therefore L = \frac{\pi}{\omega} \therefore \text{length of } (-L, L) = 2L$

Also, $(c, c + 2L) = (-L, L)$ (or) $(c, c + 2L) = \left(-\frac{\pi}{\omega}, \frac{\pi}{\omega}\right) \therefore c = \frac{\pi}{\omega}$

$c = \frac{\pi}{\omega} \quad \& \quad L = \frac{\pi}{\omega}$

----->(2)

Nature: **Neither even nor odd**

Euler's Co-efficients:

$$a_0 = \frac{1}{L} \int_{t=c}^{c+2L} f(t) dt$$



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$$\Rightarrow a_0 = \frac{1}{\pi} \int_{t=-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} f(t) dt$$

$$\Rightarrow a_0 = \frac{\omega}{\pi} \left[\int_{-\frac{\pi}{\omega}}^0 f(t) dt + \int_0^{\frac{\pi}{\omega}} f(t) dt \right]$$

$$\Rightarrow a_0 = \frac{\omega}{\pi} \left[0 + \int_0^{\frac{\pi}{\omega}} E \sin(\omega t) dt \right]$$

$$\Rightarrow a_0 = \frac{E\omega}{\pi} \int_0^{\frac{\pi}{\omega}} \sin(\omega t) dt$$

$$\Rightarrow a_0 = \frac{E\omega}{\pi} \left[\frac{-\cos(\omega t)}{\omega} \right]_0^{\frac{\pi}{\omega}}$$

$$\Rightarrow a_0 = \frac{-E\omega}{\omega\pi} \left[\frac{\cos(\omega t)}{1} \right]_0^{\frac{\pi}{\omega}} \Rightarrow a_0 = \frac{-E}{\pi} \left\{ \cos\left(\frac{\omega\pi}{\omega}\right) - \cos(0) \right\}$$

$$\Rightarrow a_0 = \frac{-E}{\pi} \{-1 - 1\} \because \cos(\pi) = -1 \text{ \& } \cos(0) = 1$$

$\therefore a_0 = \frac{2E}{\pi} \quad (or) \quad \frac{a_0}{2} = \frac{E}{\pi}$

-----> (3)

wkt

$$a_n = \frac{1}{L} \int_{t=c}^{c+2L} f(t) \cos\left(\frac{n\pi t}{L}\right) dt$$

$$\Rightarrow a_n = \frac{1}{\pi} \int_{t=-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} f(t) \cos\left(\frac{n\pi t}{\omega}\right) dt$$

$$\Rightarrow a_n = \frac{\omega}{\pi} \int_{t=-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} f(t) \cos(n\omega t) dt$$

$$\Rightarrow a_n = \frac{\omega}{\pi} \left[\int_{-\frac{\pi}{\omega}}^0 f(t) \cos(n\omega t) dt + \int_0^{\frac{\pi}{\omega}} f(t) \cos(n\omega t) dt \right]$$

$$\Rightarrow a_n = \frac{\omega}{\pi} \left[0 + \int_0^{\frac{\pi}{\omega}} E \sin(\omega t) \cos(n\omega t) dt \right]$$

$$\Rightarrow a_n = \frac{E\omega}{\pi} \left[\int_0^{\frac{\pi}{\omega}} \sin(\omega t) \cos(n\omega t) dt \right] \text{ -----> (4a)}$$



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Wkt $\sin A * \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$

$$A = \omega t \text{ \& } B = n\omega t \therefore A + B = (n + 1)\omega t \text{ \& } A - B = -(n - 1)\omega t$$

$$\sin \omega t * \cos n\omega t = \frac{1}{2} [\sin((n + 1)\omega t) + \sin(-(n - 1)\omega t)]$$

$$\therefore \sin \omega t * \cos n\omega t = \frac{1}{2} [\sin((n + 1)\omega t) - \sin((n - 1)\omega t)]$$

$$\Rightarrow a_n = \frac{E\omega}{\pi} \left[\int_0^{\frac{\pi}{\omega}} \frac{1}{2} [\sin((n + 1)\omega t) - \sin((n - 1)\omega t)] dt \right]$$

$$\Rightarrow a_n = \frac{E\omega}{2\pi} \left[\left\{ \frac{-\cos((n+1)\omega t)}{(n+1)\omega} \right\} - \left\{ \frac{-\cos((n-1)\omega t)}{(n-1)\omega} \right\} \right]_0^{\frac{\pi}{\omega}}$$

$$\Rightarrow a_n = \frac{E\omega}{2\pi\omega} \left[\left\{ \frac{-\cos((n+1)\omega t)}{(n+1)} \right\} + \left\{ \frac{\cos((n-1)\omega t)}{(n-1)} \right\} \right]_0^{\frac{\pi}{\omega}}$$

$$\Rightarrow a_n = \frac{E}{2\pi} \left[\frac{-1}{n+1} \left\{ \cos\left((n+1)\omega \frac{\pi}{\omega}\right) - \cos(0) \right\} + \right. \\ \left. + \frac{1}{n-1} \left\{ \cos\left((n-1)\omega \frac{\pi}{\omega}\right) - \cos(0) \right\} \right]$$

$$\Rightarrow a_n = \frac{E}{2\pi} \left[\frac{-1}{n+1} \{ \cos((n+1)\pi) - 1 \} + \frac{1}{n-1} \{ \cos((n-1)\pi) - 1 \} \right]$$

$$\Rightarrow a_n = \frac{E}{2\pi} \left[\frac{-1}{n+1} \{ (-1)^{n+1} - 1 \} + \frac{1}{n-1} \{ (-1)^{n-1} - 1 \} \right]$$

$$\because \cos(n\pi) = (-1)^n$$

$$\Rightarrow a_n = \frac{E}{2\pi} \left[\frac{-1}{n+1} \{ (-1)^n(-1) - 1 \} + \frac{1}{n-1} \{ (-1)^n(-1) - 1 \} \right]$$

$$\Rightarrow a_n = \frac{E}{2\pi} \left[\frac{-1(-1)}{n+1} \{ (-1)^n + 1 \} + \frac{(-1)}{n-1} \{ (-1)^n + 1 \} \right]$$



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$$\Rightarrow a_n = \frac{E\{(-1)^{n+1}\}}{2\pi} \left[\frac{1}{n+1} - \frac{1}{n-1} \right]$$

$$\Rightarrow a_n = \frac{E\{(-1)^{n+1}\}}{2\pi} \left[\frac{-2}{n^2-1} \right], n \neq 1$$

$$\Rightarrow a_n = \frac{-E}{\pi} \left[\frac{\{(-1)^{n+1}\}}{n^2-1} \right], n \neq 1$$

$$\therefore a_n = \begin{cases} -\frac{2E}{\pi} * \left(\frac{1}{n^2-1} \right), & n = \text{even} = 2, 4, 6, \dots \\ 0, & n = \text{odd} = 3, 5, 7, \dots \end{cases} \text{-----} \rightarrow (4b)$$

$$\text{When } n = 1 \text{ in eqn. (4a)} \Rightarrow a_1 = \frac{E\omega}{\pi} \left[\int_0^{\frac{\pi}{\omega}} \sin(\omega t) \cos(\omega t) dt \right]$$

$$\Rightarrow a_1 = \frac{E\omega}{\pi} \left[\int_0^{\frac{\pi}{\omega}} \frac{\sin(2\omega t)}{2} dt \right]$$

$$\Rightarrow a_1 = \frac{E\omega}{2\pi} \left[\frac{-\cos(2\omega t)}{2\omega} \right]_0^{\frac{\pi}{\omega}}$$

$$\Rightarrow a_1 = \frac{-E\omega}{4\pi\omega} [\cos(2\omega t)]_0^{\frac{\pi}{\omega}}$$

$$\Rightarrow a_1 = \frac{-E}{4\pi} \{\cos(2\pi) - \cos(0)\}$$

$$\Rightarrow a_1 = 0, \because \cos(2\pi) = \cos(0) = 1$$

$$\therefore a_1 = 0 \text{-----} \rightarrow (4c)$$

$$b_n = \frac{1}{L} \int_{t=c}^{c+2L} f(t) \sin\left(\frac{n\pi t}{L}\right) dt$$

$$\Rightarrow b_n = \frac{1}{\pi} \int_{t=-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} f(t) \sin\left(\frac{n\pi t}{\omega}\right) dt$$



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$$\Rightarrow b_n = \frac{\omega}{\pi} \int_{t=-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} f(t) \sin(n\omega t) dt$$

$$\Rightarrow b_n = \frac{\omega}{\pi} \left[\int_{-\frac{\pi}{\omega}}^0 f(t) \sin(n\omega t) dt + \int_0^{\frac{\pi}{\omega}} f(t) \sin(n\omega t) dt \right]$$

$$\Rightarrow b_n = \frac{\omega}{\pi} \left[0 + \int_0^{\frac{\pi}{\omega}} E \sin(\omega t) \sin(n\omega t) dt \right]$$

$$\Rightarrow b_n = \frac{E\omega}{\pi} \int_0^{\frac{\pi}{\omega}} \sin(\omega t) \sin(n\omega t) dt \quad \text{-----} \rightarrow (5a)$$

Wkt $\sin A * \sin B = -\frac{1}{2} [\cos(A+B) - \cos(A-B)]$

$A = \omega t$ & $B = n\omega t \therefore A+B = (n+1)\omega t$ & $A-B = -(n-1)\omega t$

$\sin \omega t * \sin n\omega t = -\frac{1}{2} [\cos(n+1)\omega t - \cos(-(n-1)\omega t)]$

$\sin(\omega t) * \sin(n\omega t) = -\frac{1}{2} [\cos((n+1)\omega t) - \cos((n-1)\omega t)] \because \cos(-\theta) = \cos(\theta)$

$$\Rightarrow b_n = -\frac{E\omega}{\pi} \int_0^{\frac{\pi}{\omega}} \frac{1}{2} [\cos((n+1)\omega t) - \cos((n-1)\omega t)] dt$$

$$\Rightarrow b_n = -\frac{E\omega}{2\pi} \left[\left\{ \frac{\sin((n+1)\omega t)}{(n+1)\omega} \right\} - \left\{ \frac{\sin((n-1)\omega t)}{(n-1)\omega} \right\} \right]_0^{\frac{\pi}{\omega}}$$

$$\Rightarrow b_n = -\frac{E\omega}{2\pi\omega} \left[\frac{1}{(n+1)} \{\sin((n+1)\omega t)\} - \frac{1}{(n-1)} \{\sin((n-1)\omega t)\} \right]_0^{\frac{\pi}{\omega}} \quad \text{-----} \rightarrow (5b)$$

$$\Rightarrow b_n = \frac{E}{2\pi} [0 - 0]$$

$$\Rightarrow b_n = 0 \because \sin(n+1)\pi = \sin(n-1)\pi = \sin 0 = 0$$

$$\Rightarrow b_n = 0, n \neq 1 \quad i.e \quad n > 1 \quad \text{-----} \rightarrow (5c)$$

$n = 1 \Rightarrow b_1 = \frac{E\omega}{\pi} \int_0^{\frac{\pi}{\omega}} \sin(\omega t) \sin(\omega t) dt$, using eqn. (5a)

$$\Rightarrow b_1 = \frac{E\omega}{\pi} \int_0^{\frac{\pi}{\omega}} \sin^2(\omega t) dt$$



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$$\text{Wkt } \sin^2(t) = \frac{1-\cos(2t)}{2} \therefore \sin^2(\omega t) = \frac{1-\cos(2\omega t)}{2}$$

$$\Rightarrow b_1 = \frac{E\omega}{\pi} \int_0^{\frac{\pi}{\omega}} \left[\frac{1-\cos(2\omega t)}{2} \right] dt$$

$$\Rightarrow b_1 = \frac{E\omega}{2\pi} \left[t - \frac{\sin(2\omega t)}{2\omega} \right]_0^{\frac{\pi}{\omega}}$$

$$\Rightarrow b_1 = \frac{E\omega}{2\pi} \left[\left\{ \frac{\pi}{\omega} - 0 \right\} - \frac{1}{2\omega} \{0 - 0\} \right] \because \sin(2\pi) = \sin(0) = 0$$

$$\Rightarrow b_1 = \frac{E\omega}{2\pi} * \frac{\pi}{\omega}$$

$$\therefore b_1 = \frac{E}{2} \quad \text{-----} \rightarrow (5d)$$

From eqns. (5c) & (5d) we have

$$\therefore b_n = \begin{cases} \frac{E}{2}, & n = 1 \\ 0, & n > 1 \end{cases} \quad \text{-----} \rightarrow (6)$$

Thus substituting the above obtained values of a_0 , a_n & b_n from eqns.(3), (4b) & (6) respectively in the Fourier series expansion of

$f(t)$ in the interval $\left(-\frac{\pi}{\omega}, \frac{\pi}{\omega}\right)$, $c = -\frac{\pi}{\omega}$ & $l = \frac{\pi}{\omega}$ we get

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi t}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi t}{l}\right), \quad c = -\frac{\pi}{\omega} \text{ \& \;} l = \frac{\pi}{\omega}$$

$$f(t) = \frac{E}{\pi} + \sum_{n=1,3,5,\dots}^{\infty} a_n \cos\left(\frac{n\pi t}{l}\right) + \sum_{n=2,4,6,\dots}^{\infty} a_n \cos\left(\frac{n\pi t}{l}\right) +$$

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$$+ b_1 \sin\left(\frac{\pi t}{l}\right) + \sum_{n=2}^{\infty} b_n \sin\left(\frac{n\pi t}{l}\right)$$

$$f(t) = \frac{E}{\pi} + 0 + \sum_{n=2,4,6,\dots}^{\infty} a_n \cos\left(\frac{n\pi t}{l}\right) + \frac{E}{2} \sin\left(\frac{\pi t}{l}\right) + 0$$

$$f(t) = \frac{E}{\pi} + \sum_{n=2,4,6,\dots}^{\infty} -\frac{2E}{\pi} * \left(\frac{1}{n^2 - 1}\right) \cos\left(\frac{n\pi t}{\omega}\right) + \frac{E}{2} \sin\left(\frac{\pi t}{\omega}\right) + 0$$

$$f(t) = \frac{E}{\pi} - \frac{2E}{\pi} \sum_{n=2,4,6,\dots}^{\infty} \left(\frac{1}{n^2 - 1}\right) \cos(n\omega t) + \frac{E}{2} \sin(\omega t)$$

$$\therefore f(t) = \frac{E}{\pi} - \frac{2E}{\pi} \left[\frac{\cos(2\omega t)}{3} + \frac{\cos(4\omega t)}{15} + \frac{\cos(6\omega t)}{35} + \dots \right] + \frac{E}{2} \sin(\omega t)$$

-----→(7)

Thus eqn.(7) gives the Fourier series of the periodic function passed through a half wave rectifier in the interval $(-L, L) = \left(-\frac{\pi}{\omega}, \frac{\pi}{\omega}\right)$

11) Obtain the Fourier series expansion of the function

$$f(x) = x^2 \text{ in the interval } \left(-\frac{3}{2}, \frac{3}{2}\right)$$

Soltn: By data, $f(x) = x^2$

-----→(1)

$$(c, c + 2l) = \left(-\frac{3}{2}, \frac{3}{2}\right) \therefore c = -\frac{3}{2} \text{ \& } l = \frac{3}{2}$$

$$\therefore c = -\frac{3}{2} \text{ \& } l = \frac{3}{2}$$

-----→(2)



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Nature of the function: Here interval is $\left(-\frac{3}{2}, \frac{3}{2}\right) = (-l, l)$

$$f(x) = x^2 \quad \therefore f(-x) = (-x)^2 \Rightarrow f(-x) = x^2 \quad \therefore f(-x) = f(x)$$

$\therefore f(x) = x^2$ is an **even** function in the interval $\left(-\frac{3}{2}, \frac{3}{2}\right)$

$$\therefore b_n = 0$$

-----→(3)

wkt when the function $f(x) = x^2$ is even in the interval $\left(-\frac{3}{2}, \frac{3}{2}\right) = (-l, l)$

$$a_0 = \frac{2}{l} * \int_{x=0}^l f(x) dx, \quad l = \frac{3}{2}$$

$$\Rightarrow a_0 = \frac{2}{3/2} * \int_{x=0}^{3/2} x^2 dx$$

$$\Rightarrow a_0 = \frac{4}{3} * \left\{ \frac{x^3}{3} \right\}_0^{3/2} \Rightarrow a_0 = \frac{4}{9} * \left\{ \left(\frac{3}{2} \right)^3 - 0 \right\}$$

$$\Rightarrow a_0 = \frac{4}{9} * \frac{27}{8} \quad \therefore a_0 = \frac{3}{2} \quad (or) \quad \frac{a_0}{2} = \frac{3}{4}$$

$$\therefore a_0 = \frac{3}{2} \quad (or) \quad \frac{a_0}{2} = \frac{3}{4}$$

-----→(4)

$$a_n = \frac{2}{l} * \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad l = \frac{3}{2}$$

$$\Rightarrow a_n = \frac{2}{(3/2)} * \int_{x=0}^{3/2} x^2 \cos\left(\frac{n\pi x}{3/2}\right) dx \Rightarrow a_n = \frac{4}{3} * \int_{x=0}^{3/2} x^2 \cos\left(\frac{2n\pi x}{3}\right) dx$$

By applying Bernoulli's rule of integration by parts,

$$u = u_0 = x^2 \quad \& \quad v = \cos\left(\frac{2n\pi x}{3}\right), \quad \int uv = u_0 v_1 - u_1 v_2 + u_2 v_3 - u_3 v_4 + \dots$$

$$\Rightarrow a_n = \frac{4}{3} * \int_{x=0}^{3/2} x^2 \cos\left(\frac{2n\pi x}{3}\right) dx$$

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$$\Rightarrow a_n = \frac{4}{3} * \left[\left\{ x^2 * \left(\frac{\sin\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)} \right) \right\} - \left\{ 2x * \left(\frac{-\cos\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^2} \right) \right\} + \left\{ 2 * \left(\frac{-\sin\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^3} \right) \right\} - 0 \right]_0^{\frac{3}{2}}$$

$$\Rightarrow a_n = \frac{4}{3} * \left[\{0 - 0\} + \left(\frac{3}{2n\pi}\right)^2 * 2 * \left\{ x * \cos\left(\frac{2n\pi x}{3}\right) \right\}_0^{\frac{3}{2}} + \{0 - 0\} \right]$$

$$\because \sin(0) = 0 = \sin(n\pi), n = 0, 1, 2, 3, 4 \dots$$

$$\Rightarrow a_n = \frac{4}{3} * \left(\frac{3}{2n\pi}\right)^2 * 2 \left[\left\{ \left(\frac{3}{2}\right) * \right\} - 0 \right]$$

$$\Rightarrow a_n = \frac{4}{3} * \left(\frac{3}{2n\pi}\right)^2 * 2 * \left(\frac{3}{2}\right) * \cos(n\pi)$$

$\therefore a_n = \left(\frac{3}{n\pi}\right)^2 * (-1)^n, n = 1, 2, 3, 4, \dots$

----->(5)

Thus substituting the above obtained values of a_0 , a_n & b_n from eqns.(4), (5) & (3) respectively in the Fourier series expansion of $f(x) = x^2$ in the interval $\left(-\frac{3}{2}, \frac{3}{2}\right)$ we get,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right), l = \frac{3}{2}$$

$$x^2 = \frac{3}{4} + \left(\frac{3}{\pi}\right)^2 \sum_{n=1}^{\infty} \frac{1}{n^2} * \cos\left(\frac{2n\pi x}{3}\right) + 0 \quad \text{and the required Fourier series is}$$

$\therefore x^2 = \frac{3}{4} + \left(\frac{3}{\pi}\right)^2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} * \cos\left(\frac{2n\pi x}{3}\right)$

----->(6)

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HALF RANGE FOURIER SERIES :

In order to express Fourier series expansion so that it contains either only sine terms or only cosine terms then the function must be defined over the interval $(0, l)$ which is half the period of the Fourier series. It is immaterial whatever may be the function outside the range/interval $(0, l)$, we extend the function to cover the range $(-l, l)$ so that the new function may be odd or even. The Fourier series expansion of such a function over the half period/interval is called as *Half Range Fourier Series* as it contains either only sine or cosine terms.

Note: The graphs for such functions of x will remain the same in the interval $(0, l)$ for different values of x but outside $(0, l)$ are different for odd or even functions.

Case(1): Sine Series

$$f(x) = \begin{cases} \phi(x) & \text{in } (0, l) \text{ as given} \\ -\phi(-x) & \text{in } (-l, 0) \text{ assumed to make } f(x) \text{ odd} \end{cases}$$

Therefore $f(x)$ becomes an **odd** function in $(-l, l)$

$$\therefore a_0 = 0 = a_n \text{ \& } b_n = \frac{2}{l} \int_{x=0}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$\therefore f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \text{ -----} \rightarrow (1)$$

The infinite series given by eqn.(1) is called **Half Range Fourier Sine Series** of $f(x)$ over the interval $(0, l)$ as it contains only sine terms.

Case(2): Cosine Series

$$f(x) = \begin{cases} \phi(x) & \text{in } (0, l) \text{ as given} \\ \phi(-x) & \text{in } (-l, 0) \text{ assumed to make } f(x) \text{ even} \end{cases}$$

Therefore $f(x)$ becomes an **even** function in $(-l, l)$

$$\therefore b_n = 0 \text{ \& } a_0 = \frac{2}{l} \int_{x=0}^l f(x) dx, \quad a_n = \frac{2}{l} \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) \text{ -----} \rightarrow (2)$$

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The infinite series given by eqn.(2) is called **Half Range Fourier Cosine Series** of $f(x)$ over the interval $(0, l)$ as it contains only cosine terms.

Similarly one can define over the interval $(0, \pi)$ as it is a special case when $l = \pi$

Summary Table:

Interval	Type of Half Range Series	$f(x)$	Fourier Co-efficients
$(0, l)$	Sine	$\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$	$b_n = \frac{2}{l} \int_{x=0}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$
$(0, l)$	Cosine	$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$	$a_0 = \frac{2}{l} \int_{x=0}^l f(x) dx$ & $a_n = \frac{2}{l} \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$
$(0, \pi)$	Sine	$\sum_{n=1}^{\infty} b_n \sin(n\pi x)$	$\frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin(n\pi x) dx$
$(0, \pi)$	Cosine	$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\pi x)$	$a_0 = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) dx$ & $a_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \cos(n\pi x) dx$

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Problems on Half Range Fourier Series:

1) Express $f(x) = x$ as a half range Fourier sine series in the interval $0 < x < \pi$.

Solt: By data, $f(x) = x \rightarrow (1) \quad x \in (0, \pi) \rightarrow (2)$

$$\text{Wkt, } b_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin(nx) dx, \quad b_n = \frac{2}{\pi} \int_{x=0}^{\pi} x \sin(nx) dx$$

By applying Bernoulli's rule of integration by parts we get

$$\Rightarrow b_n = \frac{2}{\pi} \left[\left\{ x * \left(\frac{-\cos(nx)}{n} \right) \right\} - \left\{ 1 * \left(\frac{-\sin(nx)}{n^2} \right) \right\} + 0 \right]_0^{\pi}$$

$$\Rightarrow b_n = \frac{2}{\pi} * \left[\left(\frac{-1}{n} \right) \{ (\pi * \cos(n\pi)) - 0 \} \right] \quad \because \sin(n\pi) = \sin(0) = 0$$

$$\Rightarrow b_n = \frac{-2}{\pi n} * \pi (-1)^n \quad \therefore b_n = -2 * \left\{ \frac{(-1)^n}{n} \right\}$$

$$\therefore b_n = -2 * \left\{ \frac{(-1)^n}{n} \right\}$$

----->(3)

Wkt, the expression for half range Fourier sine series expansion of $f(x)$ in $(0, l)$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \Rightarrow x = -2 * \sum_{n=1}^{\infty} \left\{ \frac{(-1)^n}{n} \right\} * \sin(nx)$$

$$x = -2 \left[\left\{ \frac{-1}{1} * \sin(x) \right\} + \left\{ \frac{1}{2} * \sin(2x) \right\} + \left\{ \frac{-1}{3} * \sin(3x) \right\} + \left\{ \frac{1}{4} * \sin(4x) \right\} + \dots \right]$$

$$x = 2 \left[\left\{ \frac{1}{1} * \sin(x) \right\} - \left\{ \frac{1}{2} * \sin(2x) \right\} + \left\{ \frac{1}{3} * \sin(3x) \right\} - \left\{ \frac{1}{4} * \sin(4x) \right\} + \dots \right]$$

Above expression gives the half range Fourier sine series of $f(x)$ in $(0, \pi)$

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2) Express $f(x) = x^2$ as a half range Fourier cosine series in the interval $0 < x < \pi$

Solt: By data $f(x) = x^2$ -----→(1) $x \in (0, \pi)$ -----→(2)

Wkt,

$$a_0 = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) dx \Rightarrow a_0 = \frac{2}{\pi} \int_{x=0}^{\pi} x^2 dx$$

$$\Rightarrow a_0 = \frac{2}{\pi} * \left[\frac{x^3}{3} \right]_0^{\pi} \Rightarrow a_0 = \frac{2}{3\pi} * \{\pi^3\} \quad \therefore a_0 = \frac{2\pi^2}{3} \quad (or) \quad \frac{a_0}{2} = \frac{\pi^2}{3}$$

$$\therefore a_0 = \frac{2\pi^2}{3} \quad (or) \quad \frac{a_0}{2} = \frac{\pi^2}{3}$$

-----→(3)

Wkt,

$$a_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \cos(\pi x) dx \quad a_n = \frac{2}{\pi} \int_{x=0}^{\pi} x^2 \cos(\pi x) dx$$

By applying Bernoulli's rule of integration by parts

$$\Rightarrow a_n = \frac{2}{\pi} \left[\left\{ x^2 * \frac{\sin(\pi x)}{\pi} \right\} - \left\{ 2x * \frac{-\cos(\pi x)}{\pi^2} \right\} + \left\{ 2 * \frac{-\sin(\pi x)}{\pi^3} \right\} - 0 \right]_0^{\pi}$$

$$\Rightarrow a_n = \frac{2}{\pi} \left[0 - \left\{ 2x * \frac{-\cos(\pi x)}{\pi^2} \right\} + 0 \right]_0^{\pi}$$

$$\therefore \sin(n\pi) = \sin(0) = 0, n = 1, 2, 3, \dots$$

$$\Rightarrow a_n = \frac{2}{\pi} * \frac{2}{\pi^2} [x * \cos(\pi x)]_0^{\pi} \Rightarrow a_n = \frac{2}{\pi} * \frac{2}{\pi^2} [\pi \cos(n\pi)]$$

$$\Rightarrow a_n = 4 * \frac{(-1)^n}{\pi^2} \quad \text{-----→(4)}$$

$$\therefore a_n = 4 * \frac{(-1)^n}{\pi^2}$$

Wkt $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\pi x) \Rightarrow x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{\pi^2} * \cos(n\pi x)$

Thus the Half Range Fourier Cosine Series expansion $f(x) = x^2$ in $0 < x < \pi$

$$\therefore x^2 = \frac{\pi^2}{3} - 4 \left[\frac{1}{1^2} \cos(x) - \frac{1}{2^2} \cos(2x) + \frac{1}{3^2} \cos(3x) - + \dots \right]$$

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- 3) Find the half range Fourier sine series of $f(x) = \begin{cases} x, & 0 < x < \pi/2 \\ \pi - x, & \pi/2 < x < \pi \end{cases}$

Soltn: Here interval $x \in (0, \pi)$ -----→(1)

$$b_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin(nx) dx \quad \Rightarrow b_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) \sin(nx) dx$$

$$\Rightarrow b_n = \frac{2}{\pi} \left[\int_{x=0}^{\pi/2} f(x) \sin(nx) dx + \int_{x=\pi/2}^{\pi} f(x) \sin(nx) dx \right]$$

$$\Rightarrow b_n = \frac{2}{\pi} \left[\int_{x=0}^{\pi/2} x \sin(nx) dx + \int_{x=\pi/2}^{\pi} (\pi - x) \sin(nx) dx \right]$$

$$\Rightarrow b_n = \frac{2}{\pi} \left[\left\{ x \cdot \left(\frac{-\cos(nx)}{n} \right) \right\} - \left\{ 1 \cdot \left(\frac{-\sin(nx)}{n^2} \right) \right\} + 0 \right]_{0}^{\pi/2}$$

$$+ \frac{2}{\pi} \left[\left\{ (\pi - x) \cdot \left(\frac{-\cos(nx)}{n} \right) \right\} - \left\{ (-1) \cdot \left(\frac{-\sin(nx)}{n^2} \right) \right\} + 0 \right]_{\pi/2}^{\pi}$$

$$\therefore b_n = \frac{4}{\pi} * \left(\frac{\sin\left(\frac{n\pi}{2}\right)}{n^2} \right) \text{ -----→(2)}$$

Wkt, the expression for half range Fourier sine series expansion of $f(x)$ in $(0, l)$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \quad \Rightarrow f(x) = \frac{4}{\pi} * \sum_{n=1}^{\infty} \left\{ \frac{1}{n^2} \right\} \sin\left(\frac{n\pi}{2}\right) * \sin(nx)$$

$$f(x) = \frac{4}{\pi} * \left\{ \sin x - \frac{\sin(3x)}{3^2} + \frac{\sin(5x)}{5^2} - + \dots \right\}$$



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4) Find the half range Fourier cosine series of $f(x) = \begin{cases} \frac{\pi}{3}, & 0 < x < \frac{\pi}{3} \\ 0, & \frac{\pi}{3} < x < \frac{2\pi}{3} \\ -\frac{\pi}{3}, & \frac{2\pi}{3} < x < \pi \end{cases}$

Soltn: Here $x \in (0, \pi)$

$$a_0 = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) dx$$

$$a_0 = \frac{2}{\pi} \left[\int_{x=0}^{\pi/3} f(x) dx + \int_{x=\pi/3}^{2\pi/3} f(x) dx + \int_{x=2\pi/3}^{\pi} f(x) dx \right]$$

$$a_0 = \frac{2}{\pi} \left[\int_{x=0}^{\pi/3} \left(\frac{\pi}{3}\right) dx + \int_{x=\pi/3}^{2\pi/3} 0 dx + \int_{x=2\pi/3}^{\pi} \left(-\frac{\pi}{3}\right) dx \right]$$

$$a_0 = \frac{2}{\pi} \left[\int_{x=0}^{\pi/3} \left(\frac{\pi}{3}\right) dx + \int_{x=\pi/3}^{2\pi/3} 0 dx + \int_{x=2\pi/3}^{\pi} \left(-\frac{\pi}{3}\right) dx \right]$$

$$a_0 = 0 \quad (\text{or}) \quad \frac{a_0}{2} = 0 \quad \text{-----} \rightarrow (1)$$

$$a_n = \frac{2}{\pi} \int_{x=0}^{\pi} f(x) * \cos(nx) dx$$

$$a_n = \frac{2}{\pi} \left[\int_{x=0}^{\pi/3} f(x) * \cos(nx) dx + \int_{x=\pi/3}^{2\pi/3} f(x) * \cos(nx) dx + \int_{x=2\pi/3}^{\pi} f(x) * \cos(nx) dx \right]$$

$$a_n = \frac{2}{\pi} \left[\left(\frac{\pi}{3}\right) \int_{x=0}^{\pi/3} \cos(nx) dx + 0 - \left(\frac{\pi}{3}\right) \int_{x=2\pi/3}^{\pi} \cos(nx) dx \right]$$

$$a_n = \frac{2}{3} \left[\frac{1}{n} \left\{ \sin\left(\frac{n\pi}{3}\right) + \sin\left(\frac{2n\pi}{3}\right) \right\} \right] \quad \text{-----} \rightarrow (2)$$

Wkt $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx)$

$$f(x) = \frac{2}{3} \sum_{n=1}^{\infty} \frac{1}{n} \left\{ \sin\left(\frac{n\pi}{3}\right) + \sin\left(\frac{2n\pi}{3}\right) \right\} \cos(nx)$$

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Exercise Problems:

1) Express $f(x) = x$ as a half range Fourier sine series in the interval $0 < x < l$.

Solt: By data, $f(x) = x \rightarrow (1) \quad x \in (0, l) \rightarrow (2)$

$$\text{Wkt, } b_n = \frac{2}{l} \int_{x=0}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx \quad \Rightarrow b_n = \frac{2}{l} \int_{x=0}^l x * \sin\left(\frac{n\pi x}{l}\right) dx,$$

By applying Bernoulli's rule of integration by parts we get

$$\Rightarrow b_n = \frac{2}{l} \left[\left\{ x * \left(\frac{-\cos\left(\frac{n\pi x}{l}\right)}{\left(\frac{n\pi}{l}\right)} \right) \right\} - \left\{ 1 * \left(\frac{-\sin\left(\frac{n\pi x}{l}\right)}{\left(\frac{n\pi}{l}\right)^2} \right) \right\} + 0 \right]_0^l$$

$$\Rightarrow b_n = \frac{2}{l} \left[\left(\frac{-l}{n\pi} \right) \left\{ x * \cos\left(\frac{n\pi x}{l}\right) \right\} + \left(\frac{l}{n\pi} \right)^2 \left\{ \sin\left(\frac{n\pi x}{l}\right) \right\} \right]_0^l$$

$$\Rightarrow b_n = \frac{2}{l} * \left[\left(\frac{-l}{n\pi} \right) \{ (l * \cos(n\pi)) - 0 \} + \left(\frac{l}{n\pi} \right)^2 \{ 0 - 0 \} \right] \quad \because \sin(n\pi) = \sin(0) = 0$$

$$\Rightarrow b_n = \frac{2}{l} * \left(\frac{-l}{n\pi} \right) * l(-1)^n \quad \therefore b_n = \frac{-2l}{\pi} * \left\{ \frac{(-1)^n}{n} \right\}$$

$$\therefore b_n = \frac{-2l}{\pi} * \left\{ \frac{(-1)^n}{n} \right\}$$

----->(3)

Wkt, the expression for half range Fourier sine series expansion of $f(x)$ in $(0, l)$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \quad \Rightarrow x = \frac{-2l}{\pi} * \sum_{n=1}^{\infty} \left\{ \frac{(-1)^n}{n} \right\} * \sin\left(\frac{n\pi x}{l}\right)$$

$$x = \frac{-2l}{\pi} \left[\left\{ \frac{-1}{1} * \sin\left(\frac{\pi x}{l}\right) \right\} + \left\{ \frac{1}{2} * \sin\left(\frac{2\pi x}{l}\right) \right\} + \left\{ \frac{-1}{3} * \sin\left(\frac{3\pi x}{l}\right) \right\} + \left\{ \frac{1}{4} * \sin\left(\frac{4\pi x}{l}\right) \right\} + \dots \right]$$

$$x = \frac{2l}{\pi} \left[\left\{ \frac{1}{1} * \sin\left(\frac{\pi x}{l}\right) \right\} - \left\{ \frac{1}{2} * \sin\left(\frac{2\pi x}{l}\right) \right\} + \left\{ \frac{1}{3} * \sin\left(\frac{3\pi x}{l}\right) \right\} - \left\{ \frac{1}{4} * \sin\left(\frac{4\pi x}{l}\right) \right\} + \dots \right]$$

Above expression gives the half range Fourier sine series of $f(x)$ in $(0, l)$



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2) Express $f(x) = x^2$ as a half range Fourier cosine series in the interval $0 < x < l$

Solt: By data $f(x) = x^2$ -----→(1) $x \in (0, l)$ -----→(2)

$$\text{Wkt, } a_0 = \frac{2}{l} \int_{x=0}^l f(x) dx \Rightarrow a_0 = \frac{2}{l} \int_{x=0}^l x^2 dx$$

$$\Rightarrow a_0 = \frac{2}{l} * \left[\frac{x^3}{3} \right]_0^l \Rightarrow a_0 = \frac{2}{3l} * \{l^3 - 0\} \quad \therefore a_0 = \frac{2l^2}{3} \quad (\text{or}) \quad \frac{a_0}{2} = \frac{l^2}{3}$$

$$\boxed{\therefore a_0 = \frac{2l^2}{3} \quad (\text{or}) \quad \frac{a_0}{2} = \frac{l^2}{3}} \quad \text{-----→(3)}$$

$$\text{Wkt, } a_n = \frac{2}{l} \int_{x=0}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx \Rightarrow a_n = \frac{2}{l} \int_{x=0}^l x^2 \cos\left(\frac{n\pi x}{l}\right) dx$$

By applying Bernoulli's rule of integration by parts

$$\Rightarrow a_n = \frac{2}{l} \left[\left\{ x^2 * \frac{\sin\left(\frac{n\pi x}{l}\right)}{\left(\frac{n\pi}{l}\right)} \right\} - \left\{ 2x * \frac{-\cos\left(\frac{n\pi x}{l}\right)}{\left(\frac{n\pi}{l}\right)^2} \right\} + \left\{ 2 * \frac{-\sin\left(\frac{n\pi x}{l}\right)}{\left(\frac{n\pi}{l}\right)^3} \right\} - 0 \right]_0^l$$

$$\Rightarrow a_n = \frac{2}{l} \left[\left(\frac{l}{n\pi} \right) \{0 - 0\} + 2 \left(\frac{l}{n\pi} \right)^2 \{l * \cos(n\pi) - 0\} - 2 \left(\frac{l}{n\pi} \right)^3 \{0 - 0\} \right]$$

$$\because \sin(n\pi) = \sin(0) = 0, n = 1, 2, 3, \dots$$

$$\Rightarrow a_n = \frac{2}{l} \left[2 \left(\frac{l}{n\pi} \right)^2 \{l * (-1)^n\} \right] \Rightarrow a_n = \frac{4}{l} * \frac{l^3}{n^2 \pi^2} * (-1)^n \Rightarrow a_n = \frac{4l^2}{n^2 \pi^2} * (-1)^n$$

$$\Rightarrow a_n = \left(\frac{2l}{\pi} \right)^2 * \frac{(-1)^n}{n^2} \quad \boxed{\therefore a_n = \left(\frac{2l}{\pi} \right)^2 * \frac{(-1)^n}{n^2}} \quad \text{-----→(4)}$$

$$\text{Wkt } f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) \Rightarrow x^2 = \frac{l^2}{3} + \left(\frac{2l}{\pi} \right)^2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} * \cos\left(\frac{n\pi x}{l}\right)$$

Thus the Half Range Fourier Cosine Series expansion $f(x) = x^2$ in $0 < x < l$

$$\boxed{\therefore x^2 = \frac{l^2}{3} - \frac{4l^2}{\pi^2} \left[\frac{1}{1^2} \cos\left(\frac{\pi x}{l}\right) - \frac{1}{2^2} \cos\left(\frac{2\pi x}{l}\right) + \frac{1}{3^2} \cos\left(\frac{3\pi x}{l}\right) - + \dots \right]}$$



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- 3) Express $f(x) = x \sin x$ as a half range Fourier cosine series in the interval
 $0 < x < \pi$

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Practice Problems on Half Range Fourier Series:

1) Express $f(x) = x^2$ as a half range Fourier sine series in the interval $0 < x < l$

Solt: $\therefore b_n = \frac{2l^2}{\pi} * \left[\left\{ \frac{(-1)^{n+1}}{n} \right\} - \frac{2}{\pi^2} \left\{ \frac{1-(-1)^n}{n^3} \right\} \right]$

$$x = \frac{2l^2}{\pi} \sum_{n=1}^{\infty} \left\{ \frac{(-1)^{n+1}}{n} \right\} - \frac{2}{\pi^2} \left\{ \frac{1-(-1)^n}{n^3} \right\} \sin \left(\frac{n\pi x}{l} \right)$$

2) Express $f(x) = x^2$ as a half range Fourier sine series in the interval $0 < x < 2$

Solt: $\therefore b_n = \frac{8}{\pi} * \left[\left\{ \frac{(-1)^{n+1}}{n} \right\} - \frac{2}{\pi^2} \left\{ \frac{1-(-1)^n}{n^3} \right\} \right]$

$$x = \frac{8}{\pi} \sum_{n=1}^{\infty} \left\{ \frac{(-1)^{n+1}}{n} \right\} - \frac{2}{\pi^2} \left\{ \frac{1-(-1)^n}{n^3} \right\} \sin \left(\frac{n\pi x}{2} \right)$$

3) Express $f(x) = x^2$ as a half range Fourier sine series in the interval $0 < x < \pi$

Solt: $\therefore b_n = \frac{2l^2}{\pi} * \left[\left\{ \frac{(-1)^{n+1}}{n} \right\} - \frac{2}{\pi^2} \left\{ \frac{1-(-1)^n}{n^3} \right\} \right]$

$$x^2 = 2\pi \sum_{n=1}^{\infty} \left[\left\{ \frac{(-1)^{n+1}}{n} \right\} - \frac{2}{\pi^2} \left\{ \frac{1-(-1)^n}{n^3} \right\} \right] \sin(nx)$$

4) Express $f(x) = x$ as a half range Fourier cosine series in the interval $0 < x < l$.

Solt: $a_0 = l$ (or) $\frac{a_0}{2} = \frac{l}{2}$, $a_n = \frac{2l}{\pi^2} * \left[\frac{(-1)^n - 1}{n^2} \right]$

$$\therefore x = \frac{l}{2} - \frac{4l}{\pi^2} \left\{ \frac{1}{1^2} \cos \left(\frac{\pi x}{l} \right) + \frac{1}{3^2} \cos \left(\frac{3\pi x}{l} \right) + \frac{1}{5^2} \cos \left(\frac{5\pi x}{l} \right) + \dots \right\}$$

5) Express $f(x) = x^2$ as a half range Fourier cosine series in the interval $0 < x < 1$

Solt: $a_0 = \frac{2}{3}$ (or) $\frac{a_0}{2} = \frac{1}{3}$, $a_n = \left(\frac{2}{\pi} \right)^2 * \frac{(-1)^n}{n^2}$

$$\therefore x^2 = \frac{1}{3} - \frac{4}{\pi^2} \left[\frac{1}{1^2} \cos(\pi x) - \frac{1}{2^2} \cos(2\pi x) + \frac{1}{3^2} \cos(3\pi x) - + \dots \right]$$

6) Express $f(x) = x^2$ as a half range Fourier cosine series in the interval $0 < x < \pi$

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Solt: $a_0 = \frac{2\pi^2}{3}$ (or) $\frac{a_0}{2} = \frac{\pi^2}{3}$, $a_n = 4 * \frac{(-1)^n}{n^2}$

$$\therefore x^2 = \frac{1}{3} - 4 \left[\frac{1}{1^2} \cos(x) - \frac{1}{2^2} \cos(2x) + \frac{1}{3^2} \cos(3x) - + \dots \right]$$

7) Express $f(x) = x \sin x$ as a half range Fourier sine series in the interval

$$0 < x < l$$

8) Express $f(x) = x \sin x$ as a half range Fourier sine series in the interval

$$0 < x < 2$$

9) Express $f(x) = x \sin x$ as a half range Fourier sine series in the interval

$$0 < x < \pi$$

10) Express $f(x) = x \sin x$ as a half range Fourier cosine series in the interval

$$0 < x < l$$

11) Express $f(x) = x \sin x$ as a half range Fourier cosine series in the interval

$$0 < x < 2$$

12) Express $f(x) = x \cos x$ as a half range Fourier cosine series in the interval

$$0 < x < l$$

13) Express $f(x) = x \cos x$ as a half range Fourier sine series in the interval

$$0 < x < l$$